Signorini's Problem

Max Stuthmann

March 19, 2026

Contents

Abstract

This report is concerned with Signorini's problem in linear elasticity, describing the equilibrium of an elastic body in unilateral contact with a rigid obstacle. Based mainly on Chapter 3 and Section 7.1 of [1], together with some Sobolev and trace theoretic material from Chapter 2, we derive and compare several equivalent formulations of the problem, including variational, strong, and mixed formulations. A particular aim is to make the underlying arguments more transparent by writing out and unfolding several proofs from the literature in greater detail in order to make them more accessible to the reader.

1 Introduction

We are given a body $\Omega \subset \mathbb{R}^d$. While all of the below may be formulated for arbitrary $d \geq 1$, it is of course reasonable to restrict ourselves to the real world application $d \in \{2, 3\}$. Further necessary topological assumptions on Ω are introduced in the next subsection. This body may be sitting on a *rigid foundation*. See Figure 1 (left). Furthermore, let our body be made of an *elastic material*, allowing Ω to deform due to *internal* and *external forces*. If these forces are large enough, Ω in its initial state as in Figure 1 (left) is not in a *balanced* configuration. The internal and external forces are not in equilibrium. If we now allow all forces to act until the body is in an equilibrium configuration, this could look like Figure 1 (right). The body Ω in this state is designated $\varphi(\Omega)$.

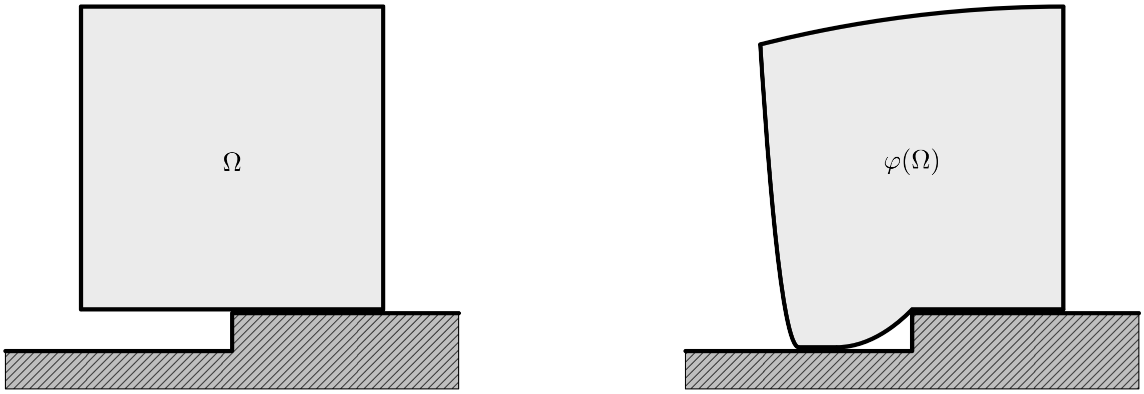

Figure 1: Body Ω on a rigid foundation in initial (left) and equilibrium (right) configuration.

The transition of Ω in its initial configuration into Ω in its equilibrium configuration $\varphi(\Omega)$ under prescribed internal and external forces defines the map

$$\varphi : \Omega \to \mathbb{R}^d, \qquad \varphi(x) = x + u(x),$$

where $u : \Omega \to \mathbb{R}^d$ is called the *displacement* function. The Signorini Problem seeks precisely this function under given elasticity properties, forces, boundary conditions and other rules.

A characteristic difficulty of the Signorini problem is that the contact conditions are neither purely of Dirichlet nor purely of Neumann type. Instead, they take the form of *unilateral* boundary conditions. The body is not allowed to *penetrate* the rigid foundation, contact forces may only be *compressive*, and both effects are coupled through a *complementarity* relation. In particular, the *effective contact zone* is not known in advance, but depends on the unknown solution itself. This makes the problem nonlinear and leads naturally to formulations in terms of variational *inequalities* and, later on, *Karush–Kuhn–Tucker* type conditions. The presentation in this report is mainly based on [1], in particular on Chapter 3 and Section 7.1, together with some Sobolev and trace theoretic material from Chapter 2.

The aim of this report is to derive and compare several equivalent formulations of the Signorini problem in a functional analytic setting. After introducing the geometrical setup, the boundary decomposition, and the required Sobolev spaces, we model the elastic response of the body and define the admissible displacement set. Based on this, we first study an energy formulation and derive the weak formulation as a variational inequality. We then pass to stronger formulations by identifying suitable normal traces on the contact boundary and recovering the contact conditions in weak and KKT-type form. Finally, we discuss a mixed formulation involving a Lagrange multiplier for the normal contact force.

2 Geometric and Physical Setup

Naturally, all of the above must be mathematically modeled in order to apply suitable theories and methods, e.g. from functional analysis. This will be covered in the following sections. Throughout, $\Omega \subset \mathbb{R}^d$, $d \in \{2,3\}$ denotes an open, bounded and connected domain with boundary $\Gamma := \partial\Omega$, that is at least Lipschitz, i.e. $\Gamma \in \mathcal{C}^{m,1}$, $m \geq 0$. Among other things, this provides us with an *outward normal*

$$\mathrm{n} \in \mathcal{C}^{m-1,1}(\Gamma;\mathbb{R}^d), \qquad m \geq 1.$$

For now, $m = 0$ is sufficient. Then, $\mathrm{n} \in L^\infty(\Gamma;\mathbb{R}^d)$. Later on, higher regularity on Γ is needed. We begin by dividing the boundary into different sub-boundaries, which will later each be subject to different boundary conditions. Then the elasticity properties of the body are appropriately modeled and we discuss how the acting forces are understood. Finally, it is defined which displacement functions are admissible at all.

2.1 Boundary Decomposition

The boundary is decomposed into three pairwise disjoint parts

$$\Gamma = \overline{\Gamma_D} \cup \overline{\Gamma_N} \cup \overline{\Gamma_C}, \qquad |\Gamma_C|, |\Gamma_D| > 0,\ |\Gamma_N| \geq 0.$$

which are assumed to be open in Γ and where each part is subject to different boundary conditions to model different physical interactions (See Figure 2).

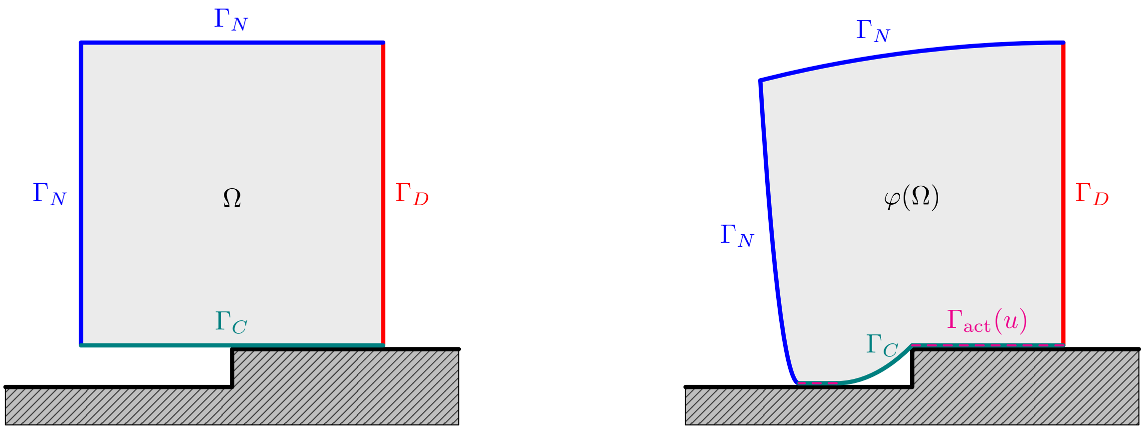

Figure 2: Decomposition of Γ in Dirichlet, Neumann and Contact parts before (left) and after displacement (right).

While Γ_C and Γ_D are assumed to be of positive $(d-1)$-Lebesgue measure, Γ_N may in some instances be a null set, or more roughly speaking, nonexistent. On Γ_D, we apply *homogeneous Dirichlet* boundary conditions. This forces the body to be *clamped* and this part, i.e. the displacement should not act on Γ_D. The part Γ_N may be subject to a *Neumann* boundary condition, which means we impose a known *surface force density*. Γ_C denotes the *potential contact* boundary. The rigid foundation is positioned so that any actual contact, both initially and in the deformed equilibrium configuration, may only occur on Γ_C. While Γ_C is prescribed a priori, the *active contact zone* $\Gamma_{\mathrm{act}}(u) \subset \Gamma_C$ is unknown and depends on the solution u. Later on, the boundary conditions imposed on Γ_C enforce *unilateral contact* through *nonpenetration* and *complementarity*. The rigid foundation can be modeled through a *gap-function* $g : \Gamma_C \to \mathbb{R}$, describing the distance between Γ_C and the foundation. While the foundation in Figure 2 suggests a piecewise constant gap-function g, we will throughout the whole report assume $g = 0$.

2.2 Basic Notations on Sobolev Spaces

This subsection is intended to fix some of the notations used in this report. Since Signorini's problem can be formulated as a partial differential equation for the displacement function, Sobolev spaces are of particular importance. The Sobolev space $W^{k,p}(\Omega;\mathbb{R}^d)$ contains every $L^p(\Omega;\mathbb{R}^d)$-function, such that weak derivatives up to order $k \geq 0$ are again in $L^p(\Omega;\mathbb{R}^d)$. Here, we are only interested in the Hilbert space case $p = 2$ and we write

$$H^k(\Omega;\mathbb{R}^d) := W^{k,2}(\Omega;\mathbb{R}^d).$$

The L^2-inner product and norm is denoted by $(\cdot,\,\cdot)_{0,\Omega}$ and $\|\cdot\|_{0,\Omega}$. We record this introduction in the following definition and simultaneously establish a notation for the associated norm.

Definition 2.1 (Sobolev Spaces of Integer Order)**.** *Let $k \in \mathbb{N}_0$. The* `Sobolev space` $H^k(\Omega;\mathbb{R}^d)$ *is defined by*

$$H^k(\Omega;\mathbb{R}^d) := \left\{ u \in L^2(\Omega;\mathbb{R}^d) \,\middle|\, \mathrm{D}^\alpha u \in L^2(\Omega;\mathbb{R}^d)\ \forall\, \alpha \in \mathbb{N}_0^d,\ |\alpha| \leq k \right\},$$

where $\mathrm{D}^\alpha u$ is understood in the weak sense. Moreover, we equip $H^k(\Omega;\mathbb{R}^d)$ with the norm

$$\|u\|_{k,\Omega} := \left(\sum_{|\alpha|\leq k} \|\mathrm{D}^\alpha u\|_{0,\Omega}^2 \right)^{1/2}, \qquad u \in H^k(\Omega;\mathbb{R}^d).$$

We adopt the usual convention $H^0(\Omega;\mathbb{R}^d) = L^2(\Omega;\mathbb{R}^d)$ explaining the notion of the L^2-norm. Furthermore, we state that $H^k(\Omega;\mathbb{R}^d)$ is a Hilbert space.

Proposition 2.1 ($H^k(\Omega;\mathbb{R}^d)$ is a Hilbert Space)**.** *The Sobolev space $H^k(\Omega;\mathbb{R}^d)$ together with the inner product*

$$(u,\, v)_{k,\Omega} := \sum_{|\alpha|\leq k} (\mathrm{D}^\alpha u,\, \mathrm{D}^\alpha v)_{0,\Omega}, \qquad u, v \in H^k(\Omega;\mathbb{R}^d)$$

is a Hilbert space with induced norm $\|\cdot\|_{k,\Omega}$. □

After the integer-order spaces $H^k(\Omega;\mathbb{R}^d)$, we also need Sobolev spaces of non-integer order. The main reason is that trace operators naturally lead to boundary spaces of fractional order. In particular, traces of $H^1(\Omega;\mathbb{R}^d)$-functions are, in general, not measured in another integer-order Sobolev space, but in a space of order $1/2$ on the boundary and similarly on boundary parts $\Gamma_0 \subset \Gamma$. For a non-integer order

$$s = k + \theta, \qquad k \in \mathbb{N},\ \theta \in (0,1),$$

the remaining fractional part θ is measured by difference quotients. More precisely, one measures how strongly a function oscillates between pairs of points $x, y \in \Omega$, with a weight that becomes singular as $|x - y| \to 0$. This leads to the following quantity.

Definition 2.2 (Slobodeckij Seminorm)**.** *Let $\theta \in (0,1)$. For a measurable function $u : \Omega \to \mathbb{R}^d$, we define the* `Slobodeckij semi-norm`

$$|u|_{\theta,\Omega} := \left(\int_\Omega \int_\Omega \frac{|u(x) - u(y)|^2}{|x - y|^{d+2\theta}}\, \mathrm{d}x\, \mathrm{d}y \right)^{1/2},$$

whenever the right-hand side is finite.

The quantity $|u|_{\theta,\Omega}$ is only a semi-norm, since it vanishes on constant functions. Combined with an L^2-term, however, it yields a genuine norm. This allows us to define Sobolev spaces of fractional order and, as above, to equip them with a Hilbert space structure.

Definition 2.3 (Sobolev Spaces of Fractional Order)**.** *Let $s = k+\theta$ with $k \in \mathbb{N}_0$ and $\theta \in (0,1)$. The* `Sobolev space` *of fractional order s is defined by*

$$H^s(\Omega;\mathbb{R}^d) := \left\{ u \in H^k(\Omega;\mathbb{R}^d) \;\middle|\; |\mathrm{D}^\alpha u|_{\theta,\Omega} < \infty \;\forall\, \alpha \in \mathbb{N}_0^d,\; |\alpha| = k \right\},$$

where $\mathrm{D}^\alpha u$ is understood in the weak sense. Moreover, we define

$$|u|_{s,\Omega} := \left(\sum_{|\alpha|=k} |\mathrm{D}^\alpha u|^2_{\theta,\Omega} \right)^{1/2}, \qquad u \in H^s(\Omega;\mathbb{R}^d),$$

and equip $H^s(\Omega;\mathbb{R}^d)$ with the norm

$$\|u\|_{s,\Omega} := \left(\|u\|^2_{k,\Omega} + |u|^2_{s,\Omega} \right)^{1/2}, \qquad u \in H^s(\Omega;\mathbb{R}^d).$$

Proposition 2.2 ($H^s(\Omega;\mathbb{R}^d)$ is a Hilbert Space)**.** *Let $s = k + \theta$ with $k \in \mathbb{N}_0$ and $\theta \in (0,1)$. Then $H^s(\Omega;\mathbb{R}^d)$ is a Hilbert space with inner product*

$$(u,\, v)_{s,\Omega} := (u,\, v)_{k,\Omega} + \sum_{|\alpha|=k} \int_\Omega \int_\Omega \frac{(\mathrm{D}^\alpha u(x) - \mathrm{D}^\alpha u(y)) \cdot (\mathrm{D}^\alpha v(x) - \mathrm{D}^\alpha v(y))}{|x-y|^{d+2\theta}} \,\mathrm{d}x\,\mathrm{d}y$$

for $u, v \in H^s(\Omega;\mathbb{R}^d)$. The induced norm is $\|\cdot\|_{s,\Omega}$. □

For $u \in H^s(\Omega;\mathbb{R}^d)$, pointwise boundary values are in general not defined. The trace theorem provides a canonical replacement.

Definition 2.4 (Classical Trace on Smooth Functions)**.** *We define the* `classical trace operator`

$$\gamma_0 : \mathcal{C}^\infty(\overline{\Omega};\mathbb{R}^d) \to L^2(\Gamma;\mathbb{R}^d), \qquad \gamma_0 u := u|_\Gamma$$

on the space

$$\mathcal{C}^\infty(\overline{\Omega};\mathbb{R}^d) := \left\{ u|_{\overline{\Omega}} \;\middle|\; u \in \mathcal{C}^\infty(U;\mathbb{R}^d) \text{ for some open neighborhood } U \supset \overline{\Omega} \right\}.$$

Definition 2.5 (Boundary Sobolev Spaces)**.** *Let $s > 1/2$. We define*

$$H^{s-1/2}(\Gamma;\mathbb{R}^d) := \gamma_0\big(\mathcal{C}^\infty(\overline{\Omega};\mathbb{R}^d)\big)$$

equipped with the norm

$$\|g\|_{s-\frac{1}{2},\Gamma} := \inf\left\{ \|u\|_{s,\Omega} \;\middle|\; u \in \mathcal{C}^\infty(\overline{\Omega};\mathbb{R}^d),\; \gamma_0 u = g \right\}.$$

Notice how the underlying set is the same for different values of $s > 1/2$, but it is equipped with different norms. The classical trace operator now extends uniquely to a continuous trace operator on $H^s(\Omega;\mathbb{R}^d)$.

Theorem 2.1 (Trace Theorem)**.** *Let $\Gamma \in \mathcal{C}^{m,1}$, $m \geq 0$ and $1/2 < s \leq m+1$. Then the classical trace operator γ_0 extends uniquely to a continuous surjective linear operator*

$$\gamma : H^s(\Omega; \mathbb{R}^d) \to H^{s-1/2}(\Gamma; \mathbb{R}^d).$$

In particular, there exists a constant $C_{\mathrm{tr}} > 0$ such that

$$\|\gamma u\|_{s-\frac{1}{2},\Gamma} \leq C_{\mathrm{tr}} \|u\|_{s,\Omega} \qquad \forall\, u \in H^s(\Omega; \mathbb{R}^d).$$

$\square$

In the special case $s = 1$, we further have

$$\ker(\gamma) = \overline{\mathcal{D}(\Omega; \mathbb{R}^d)}^{\|\cdot\|_{1,\Omega}} =: H_0^1(\Omega; \mathbb{R}^d),$$

where $\mathcal{D}(\Omega; \mathbb{R}^d)$ denotes the space of $\mathbb{R}^d$-valued test functions on Ω.

2.3 Elasticity and Forces

In elasticity, stresses are described by *symmetric second order tensors*,

$$\mathbb{S}^d := \left\{ s \in \mathbb{R}^d \otimes \mathbb{R}^d \;\middle|\; s^\top = s \right\}.$$

We identify the tensor space by symmetric $d \times d$-matrices in the canonical basis. Indeed,

$$\mathbb{S}^d \cong \left\{ S \in \mathbb{R}^{d \times d} \;\middle|\; S^\top = S \right\}.$$

As an inner product on $\mathbb{R}^{d \times d}$, we use the standard Frobenius product

$$A : B := \sum_{j=1}^{d} \sum_{k=1}^{d} A_{jk}\, B_{jk}, \qquad |A| := \sqrt{A : A}$$

for real $d \times d$-matrices A and B, while for $x, y \in \mathbb{R}^d$ we denote by

$$x \cdot y := \sum_{j=1}^{d} x_j\, y_j, \qquad |x| := \sqrt{x \cdot y},$$

the Euclidean inner product and norm. As $\mathbb{S}^d$ may be identified with a subspace of $\mathbb{R}^{d^2}$, the above definitions of Sobolev spaces may be naturally extended to $\mathbb{S}^d$-valued functions on Ω. We continue by defining the *small strain tensor* for a displacement function $u \in H^1(\Omega; \mathbb{R}^d)$, which is similar to the symmetric part of square matrices but for the gradient ∇u of u and through which the deformation of the body $\overline{\Omega}$ by u is quantified.

Definition 2.6 (Small Strain Tensor)**.** *Let $s \geq 1$. The* `(small) strain tensor` *ε is defined by*

$$\varepsilon : H^s(\Omega; \mathbb{R}^d) \to H^{s-1}(\Omega; \mathbb{S}^d), \qquad u \mapsto \varepsilon(u) := \frac{1}{2}\left(\nabla u + \nabla u^\top\right).$$

In linear elasticity, stresses are assumed to depend linearly on strains. The material response is encoded in the fourth order *elasticity tensor* $\mathcal{A}$, which maps a symmetric strain tensor to the corresponding *Cauchy stress tensor*. The tensor field $\mathcal{A}$ is prescribed by the material and may vary in space, describing a *heterogeneous* body. For a fourth order tensor $\mathcal{A} : \Omega \to \mathcal{L}(\mathbb{S}^d; \mathbb{S}^d)$ and $s \in \mathbb{S}^d$, we use the *double contraction*

$$(\mathcal{A}(x) : s)_{ij} := \sum_{k=1}^{d} \sum_{l=1}^{d} \mathcal{A}_{ijkl}(x)\, s_{kl}, \qquad s : \mathcal{A}(x) : s := \sum_{i=1}^{d} \sum_{j=1}^{d} \sum_{k=1}^{d} \sum_{l=1}^{d} \mathcal{A}_{ijkl}(x)\, s_{ij}\, s_{kl}.$$

Definition 2.7 (Elasticity Tensor Field). *An* `elasticity tensor field` *is a fourth order tensor*

$$\mathcal{A} : \Omega \to \mathcal{L}(\mathbb{S}^d; \mathbb{S}^d)$$

such that the coefficients satisfy the symmetries

$$\mathcal{A}_{ijkl}(x) = \mathcal{A}_{jikl}(x) = \mathcal{A}_{ijlk}(x) = \mathcal{A}_{klij}(x), \qquad \forall\, 1 \le i, j, k, l \le d,$$

and the uniform ellipticity condition, i.e. there exists $\alpha_{\mathcal{A}} > 0$ such that for almost every $x \in \Omega$

$$s : \mathcal{A}(x) : s \ge \alpha_{\mathcal{A}}\,(s : s) \qquad \forall\, s \in \mathbb{S}^d.$$

Moreover, there exists $C_{\mathcal{A}} > 0$ such that for a.e. $x \in \Omega$

$$|\mathcal{A}_{ijkl}(x)| \le C_{\mathcal{A}} \qquad \forall\, 1 \le i, j, k, l \le d.$$

The *internal* and *surface forces* in the body $\overline{\Omega}$ are described by the *Cauchy stress tensor*.

Definition 2.8 (Cauchy Stress Tensor). *Let $s \ge 1$. We define the* `(linear) Cauchy stress tensor` *by*

$$\sigma : H^s(\Omega; \mathbb{R}^d) \to H^{s-1}(\Omega; \mathbb{S}^d), \qquad u \mapsto \mathcal{A} : \varepsilon(u),$$

where $\mathcal{A}$ is a sufficiently smooth elasticity tensor field, such that σ is well-defined.

In the case $s = 1$, it is sufficient to require $\mathcal{A}_{ijkl} \in L^\infty(\Omega)$ for every $1 \le i, j, k, l \le d$ for $\sigma(u) = \mathcal{A} : \varepsilon(u)$ to be part of $L^2(\Omega; \mathbb{S}^d)$ for every $u \in H^1(\Omega; \mathbb{R}^d)$. Later on, higher order Sobolev spaces are asked for, which requires higher regularity of $\mathcal{A}$. For example, for $3/2 < s \le 2$ it is sufficient if $\mathcal{A}_{ijkl} \in W^{1,\infty}(\Omega)$ for every $1 \le i, j, k, l \le d$, while constant $\mathcal{A}$ always lead to well-defined stress tensors σ. The function $f \in L^2(\Omega; \mathbb{R}^d)$ may describe a *volume force density* and $F \in L^2(\Gamma_N; \mathbb{R}^d)$ may represent *surface loads* on the Neumann boundary Γ_N. With the basic principle that work is force times displacement, it is reasonable to define the linear functional

$$\ell(v) := \int_\Omega f \cdot v \,\mathrm{d}x + \int_{\Gamma_N} F \cdot (\gamma v) \,\mathrm{d}s, \qquad v \in H^1(\Omega; \mathbb{R}^d), \tag{2.1}$$

describing the *virtual work of external forces*. As the internal forces are described by the stress $\sigma(u) = \mathcal{A} : \varepsilon(u)$, the same principle suggests the symmetric bilinear form

$$a(u, v) := \int_\Omega \sigma(u) : \varepsilon(v) \,\mathrm{d}x, \qquad u, v \in H^1(\Omega; \mathbb{R}^d). \tag{2.2}$$

as a representative for the *virtual work of internal forces*. Equilibrium between both reads $a(u, v) = \ell(v)$ for all $v \in H^1(\Omega; \mathbb{R}^d)$, which already looks like a weak formulation but at this point, only the Neumann boundary conditions on Γ_N are imposed.

2.4 Admissible Displacements

We start by introducing the underlying Hilbert space V in which homogeneous Dirichlet boundary conditions on Γ_D are already incorporated by

$$V := \left\{ v \in H^1(\Omega; \mathbb{R}^d) \;\middle|\; (\gamma v)|_{\Gamma_D} = 0 \right\} \subset H^1(\Omega; \mathbb{R}^d).$$

For $v \in V$ we define the *normal component* of the boundary displacement on Γ by $v_{\mathrm{n}} := (\gamma v) \cdot \mathrm{n}$. Here, since Γ is Lipschitz, $\Gamma \in \mathcal{C}^{0,1}$, the outward normal n belongs to $L^\infty(\Gamma; \mathbb{R}^d)$ which yields $v_{\mathrm{n}} \in L^2(\Gamma)$. The rigid foundation is located on the exterior side of Γ_C in direction of n. Hence the body is not allowed to move across Γ_C into the obstacle. This yields that the solution must satisfy the *nonpenetration* condition

$$u_{\mathrm{n}} \leq 0 \qquad \text{a.e. on } \Gamma_C. \tag{2.3}$$

and motivates the nonempty, closed and convex cone of *admissible displacements*

$$K := \{v \in V \mid v_{\mathrm{n}} \leq 0 \text{ a.e. on } \Gamma_C\} \subset V. \tag{2.4}$$

Proposition 2.3 (K is a closed convex cone)**.** *The set K defined in 2.4 is a nonempty, closed and convex cone in V.*

Proof. K is nonempty, since $0 \in K$. Let $\lambda \in [0, \infty)$ and $v \in K \subset V$. Since V is a vector space, we have $\lambda v \in V$. Furthermore,

$$(\lambda v)_{\mathrm{n}} = \gamma(\lambda v) \cdot \mathrm{n} = (\lambda \gamma v) \cdot \mathrm{n} = \lambda(\gamma v \cdot \mathrm{n}) = \lambda v_{\mathrm{n}}.$$

Since $v_{\mathrm{n}} \leq 0$ a.e. on Γ_C and $\lambda \geq 0$, we also have $\lambda v_{\mathrm{n}} \leq 0$ a.e. on Γ_C, i.e. $\lambda v \in K$. Thus K is a cone in V. Let $v, w \in K$ and $\theta \in [0, 1]$ and set $u := \theta v + (1 - \theta)w \in V$. Then we have

$$u_{\mathrm{n}} = (\gamma u) \cdot \mathrm{n} = (\theta \gamma v + (1 - \theta)\gamma w) \cdot \mathrm{n} = \theta(\gamma v \cdot \mathrm{n}) + (1 - \theta)(\gamma w \cdot \mathrm{n}) = \theta v_{\mathrm{n}} + (1 - \theta) w_{\mathrm{n}}$$

and $v_{\mathrm{n}}, w_{\mathrm{n}} \leq 0$ a.e. on Γ_C and $\theta, (1 - \theta) \geq 0$. Thus,

$$\theta v_{\mathrm{n}} + (1 - \theta) w_{\mathrm{n}} \leq 0 \qquad \text{a.e. on } \Gamma_C,$$

i.e. $u \in K$, which makes K convex. To prove that K is closed, let $v_k \in K$ and assume that $v_k \to v$ in V. Since the trace operator $\gamma : V \to H^{1/2}(\Gamma; \mathbb{R}^d)$ is continuous and $\mathrm{n} \in L^\infty(\Gamma; \mathbb{R}^d)$, the mapping

$$T : V \to L^2(\Gamma_C), \qquad T(w) := ((\gamma w) \cdot \mathrm{n})|_{\Gamma_C},$$

is continuous. Hence,

$$T(v_k) \to T(v) \qquad \text{in } L^2(\Gamma_C).$$

Since $T(v_k) \leq 0$ almost everywhere on Γ_C for all k, there exists a subsequence $T(v_{k_j})$ such that

$$T(v_{k_j})(x) \to T(v)(x) \qquad \text{for a.e. } x \in \Gamma_C.$$

Passing to the limit in the inequality $T(v_{k_j})(x) \leq 0$, we obtain

$$T(v)(x) \leq 0 \qquad \text{for a.e. } x \in \Gamma_C.$$

Thus $v \in K$, and therefore K is closed. ■

In the following, we want to prove standard properties of a and ℓ on V that are needed later on. For this, we will need the following inequalities. The first inequality is a result of *Korn's second inequality* ([1, Theorem 3.3] and [1, Section 3.5.2]) and the *Peetre–Tartar* Lemma ([1, Lemma 3.5]). Therefore only the proof of the second one is given.

Lemma 2.1. *There exists a constant $C_K > 0$ such that*

$$C_K^{-1} \|v\|_{1,\Omega} \leq \|\varepsilon(v)\|_{0,\Omega} \leq \|v\|_{1,\Omega} \qquad \forall\, v \in V.$$

Proof. We prove the second inequality. For all $v \in V$ we have

$$|\varepsilon(v)(x)| = \left|\frac{1}{2}\left(\nabla v(x) + \nabla v^\top\right)\right| \leq \frac{1}{2}\left(|\nabla v(x)| + |\nabla v(x)^\top|\right) = |\nabla v(x)|$$

and thus

$$\|\varepsilon(v)\|_{0,\Omega}^2 = \int_\Omega |\varepsilon(v)(x)|^2 \, \mathrm{d}x \leq \int_\Omega |\nabla v(x)|^2 \, \mathrm{d}x = \|\nabla v\|_{0,\Omega}^2 \leq \|v\|_{1,\Omega}^2,$$

where the last inequality holds due to

$$\|v\|_{1,\Omega}^2 = \|v\|_{0,\Omega}^2 + \|\nabla v\|_{0,\Omega}^2.$$

■

Proposition 2.4. *The symmetric bilinear form $a : V \times V \to \mathbb{R}$ defined in (2.2) is*

i) bounded, *i.e. for all $u, v \in V$ there exists $C_1 < \infty$ such that*

$$|a(u,v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega} \qquad \textit{with} \qquad C_1 = C_{\mathcal{A}},$$

ii) V-elliptic, *i.e. for all $v \in V \setminus \{0\}$ there exist $C_2 > 0$ such that*

$$a(v,v) \geq C_2 \|v\|_{1,\Omega}^2 \qquad \textit{with} \qquad C_2 = \frac{\alpha_{\mathcal{A}}}{C_K^2}.$$

Moreover, the linear functional ℓ defined in (2.1) is bounded, *i.e. for all $v \in V$ there exists $C_3 > 0$ such that*

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega} \qquad \textit{with} \qquad C_3 = \max\{1, C_{\mathrm{tr}}\} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right) \tag{2.5}$$

Proof. Let $v, w \in V$ be arbitrary. Then

$$|a(v,w)| \leq \|\mathcal{A} : \varepsilon(v)\|_{0,\Omega} \|\varepsilon(w)\|_{0,\Omega}$$

by the Cauchy–Schwarz inequality. With Definition 2.7 there exists $C_{\mathcal{A}} > 0$ such that

$$\|\mathcal{A} : \varepsilon(v)\|_{0,\Omega} \leq C_{\mathcal{A}} \|\varepsilon(v)\|_{0,\Omega}.$$

Combining this with the second inequality form Lemma 2.1 we have

$$|a(v,w)| \leq C_{\mathcal{A}} \|\varepsilon(v)\|_{0,\Omega} \|\varepsilon(w)\|_{0,\Omega} \leq C_{\mathcal{A}} \|v\|_{1,\Omega} \|w\|_{1,\Omega}.$$

The ellipticity of a follows by

$$a(v,v) = \int_\Omega \varepsilon(v) : \mathcal{A} : \varepsilon(v) \, \mathrm{d}x \geq \alpha_{\mathcal{A}} \int_\Omega \varepsilon(v) : \varepsilon(v) = \alpha_{\mathcal{A}} \|\varepsilon(v)\|_{0,\Omega}^2,$$

where $\alpha_{\mathcal{A}} > 0$ exists due to Definition 2.7. The first inequality of Lemma 2.1 finally yields

$$a(v,v) \geq \alpha_{\mathcal{A}} \|\varepsilon(v)\|_{0,\Omega}^2 \geq \alpha_{\mathcal{A}} \left(C_K^{-1} \|v\|_{1,\Omega}\right)^2 = \frac{\alpha_{\mathcal{A}}}{C_K^2} \|v\|_{1,\Omega}^2.$$

Again by Cauchy–Schwarz inequality,

$$|\ell(v)| \leq \|f\|_{0,\Omega} \|v\|_{0,\Omega} + \|F\|_{0,\Gamma_N} \|\gamma v\|_{0,\Gamma_N} \qquad \forall\, v \in V.$$

By the Trace Theorem 2.1 there exists $C_{\mathrm{tr}} > 0$ such that

$$\|\gamma v\|_{0,\Gamma_N} \leq \|\gamma v\|_{0,\Gamma} \leq \|\gamma v\|_{\frac{1}{2},\Gamma} \leq C_{\mathrm{tr}} \|v\|_{1,\Omega} \qquad \forall\, v \in V.$$

Finally,

$$|\ell(v)| \leq \left(\|f\|_{0,\Omega} + C_{\mathrm{tr}} \|F\|_{0,\Gamma_N}\right) \|v\|_{1,\Omega} \leq \max\{1, C_{\mathrm{tr}}\} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right) \|v\|_{1,\Omega}$$

for every $v \in V$, which proves (2.5). ■

3 Energy Minimization and Well-Posedness

As stated earlier in Section 2.3, equilibrium between the virtual work of internal and external forces suggests to look for $u \in H^1(\Omega;\mathbb{R}^d)$ such that $a(u,v) = \ell(v)$ for all $v \in H^1(\Omega;\mathbb{R}^d)$. Here, only the Neumann conditions are imposed. Thus, a restriction on V, which imposes the homogeneous Dirichlet boundary conditions on Γ_D and further a restriction on the closed, convex cone K of displacements $v \in V$ that fulfil the nonpenetration condition on Γ_C, is reasonable. Naturally, this yields the quadratic functional

$$\mathcal{J}(v) := \frac{1}{2}\,a(v,v) - \ell(v) = \frac{1}{2}\int_\Omega \sigma(v) : \varepsilon(v)\,\mathrm{d}x - \left[\int_\Omega f \cdot v\,\mathrm{d}x + \int_{\Gamma_N} F \cdot (\gamma v)\,\mathrm{d}s\right] \tag{3.1}$$

on V describing the *potential energy* of the elastic body to be minimized over $K \subset V$ to minimize the difference between the virtual work of internal and external forces under the nonpenetration constraint. This is already the first formulation of Signorini's problem as the minimization problem of $\mathcal{J}$ over K.

Problem 3.1 (Signorini's Problem – Energy Minimization)**.**

$$\text{Find } u \in K \text{ such that:} \qquad u \in \operatorname*{arg\,min}_{v \in K} \mathcal{J}(v).$$

Here we have decided to note u as an element of those v that minimize $\mathcal{J}$ over K. The reason for this is the lack of evidence so far of the problem's well-posedness, which includes existence and uniqueness of a minimizer and stability in the sense of a continuous dependence on the given data. Once proven, the inclusion can be replaced by an equality. Thus, we now want to prove well-posedness of the above Problem 3.1. Every formulation of Signorini's problem, which is proven to be equivalent to the above formulation is then also proven to be well-posed. We start with the uniqueness, which follows directly by the standard arguments of strict convexity of $\mathcal{J}$ and the convexity of K. The latter was already proven in Proposition 2.3.

Lemma 3.1 (Strict Convexity of $\mathcal{J}$)**.** *The functional $\mathcal{J}$ defined in (3.1) is strictly convex.*

Proof. For $u, v \in V$ and $t \in [0,1]$, we set $w_t := (1-t)u + tv$. Then

$$a(w_t, w_t) = (1-t)\,a(u,u) + t\,a(v,v) - t(1-t)\,a(u-v, u-v)$$

and

$$\ell(w_t) = (1-t)\,\ell(u) + t\,\ell(v).$$

Together, we have

$$\mathcal{J}(w_t) = (1-t)\,\mathcal{J}(u) + t\,\mathcal{J}(v) - \frac{1}{2}t(1-t)\,a(u-v, u-v).$$

If $u \neq v$ then $u - v \neq 0$ and thus by V-ellipticity of a we have $a(u-v, u-v) > 0$ and $t(1-t) > 0$. This implies

$$\mathcal{J}(w_t) < (1-t)\,\mathcal{J}(u) + t\,\mathcal{J}(v).$$

■

Proposition 3.1 (Uniqueness of Minimizer of $\mathcal{J}$)**.** *If there exists a minimizer of the functional $\mathcal{J}$ defined in (3.1), then it is unique.*

Proof. Assume that there exist two minimizers $u_1, u_2 \in K$ of $\mathcal{J}$ on K, i.e.

$$\mathcal{J}(u_1) = \min_{v \in K} \mathcal{J}(v) = \mathcal{J}(u_2). \tag{3.2}$$

If $u_1 = u_2$ there is nothing to show. Hence assume $u_1 \neq u_2$. Since K is convex, for every $t \in (0,1)$ the convex combination

$$w_t := (1-t)u_1 + tu_2$$

belongs to K. By Lemma 3.1, $\mathcal{J}$ is strictly convex and therefore we have

$$\mathcal{J}(w_t) < (1-t)\mathcal{J}(u_1) + t\mathcal{J}(u_2) \qquad \forall\, t \in (0,1).$$

Using (3.2), this yields

$$\mathcal{J}(w_t) < (1-t)\min_{v\in K}\mathcal{J}(v) + t\min_{v\in K}\mathcal{J}(v) = \min_{v\in K}\mathcal{J}(v).$$

This contradicts the definition of the minimum, since $w_t \in K$. Therefore the assumption $u_1 \neq u_2$ is false and we conclude $u_1 = u_2$. ∎

It remains to show the existence of at least one minimizer. For this, we verify *coercivity* and *weakly lower semicontinuity* of the functional $\mathcal{J}$. The proof requires the following lemma. Weak convergence is denoted by $\rightharpoonup$.

Lemma 3.2 (Weak lower semicontinuity of the norm)**.** *Let $(X, \|\cdot\|_X)$ be a normed space. Then $\|\cdot\|_X : X \to \mathbb{R}_{\geq 0}$ is a weakly lower semicontinuous functional, i.e. if $x_j \rightharpoonup x$ in X, then*

$$\|x\|_X \leq \liminf_{j\to\infty} \|x_j\|_X.$$

Proof. Let $J : X \to X^{**}$ be the canonical embedding defined by

$$(Jx)(\varphi) := \varphi(x) \qquad \forall\, \varphi \in X'.$$

By the Hahn–Banach theorem, J is an isometry, i.e. $\|Jx\|_{X^{**}} = \|x\|_X$ for all $x \in X$. Therefore,

$$\|x\|_X = \|Jx\|_{X^{**}} = \sup_{\substack{\varphi\in X' \\ \|\varphi\|_{X'}\leq 1}} |(Jx)(\varphi)| = \sup_{\substack{\varphi\in X' \\ \|\varphi\|_{X'}\leq 1}} |\varphi(x)|.$$

If $x_j \rightharpoonup x$ in X, then $\varphi(x_j) \to \varphi(x)$ for all $\varphi \in X'$ and hence

$$\|x\|_X = \sup_{\substack{\varphi\in X' \\ \|\varphi\|_{X'}\leq 1}} |\varphi(x)| \leq \liminf_{j\to\infty} \sup_{\substack{\varphi\in X' \\ \|\varphi\|_{X'}\leq 1}} |\varphi(x_j)| = \liminf_{j\to\infty} \|x_j\|_X.$$

∎

Lemma 3.3 ($\mathcal{J}$ is coercive and w.l.s.c.)**.** *The functional $\mathcal{J}$ defined in (3.1) is*

i) coercive *on V, i.e.*

$$\mathcal{J}(v) \to \infty \qquad \textit{as} \qquad \|v\|_{1,\Omega} \to \infty,$$

ii) weakly lower semicontinuous *on V, i.e. if $v_j \rightharpoonup v$ in V, then*

$$\mathcal{J}(v) \leq \liminf_{j\to\infty} \mathcal{J}(v_j).$$

Proof. By boundedness of ℓ we have $|\ell(v)| \leq \|\ell\|_{V'}\,\|v\|_{1,\Omega}$. Hence with the V-ellipticity of a, we have

$$\mathcal{J}(v) = \frac{1}{2}\,a(v,v) - \ell(v) \geq \frac{\alpha_{\mathcal{A}}}{2C_K^2}\,\|v\|_{1,\Omega}^2 - \|\ell\|_{V'}\,\|v\|_{1,\Omega} \xrightarrow{\|v\|_{1,\Omega}\to\infty} \infty.$$

For $s \in L^2(\Omega;\mathbb{S}^d)$ set

$$\|s\|_{\mathcal{A}}^2 := \int_\Omega s : \mathcal{A} : s \,\mathrm{d}x.$$

By uniform ellipticity of $\mathcal{A}$, the map $\|\cdot\|_{\mathcal{A}}$ defines a norm on $L^2(\Omega;\mathbb{S}^d)$ equivalent to $\|\cdot\|_{0,\Omega}$. Moreover, $a(v,v) = \|\varepsilon(v)\|_{\mathcal{A}}^2$. The mapping $v \mapsto \varepsilon(v)$ is linear and bounded from V to $L^2(\Omega;\mathbb{S}^d)$. Hence, if $v_j \rightharpoonup v$ in V, then

$$\varepsilon(v_j) \rightharpoonup \varepsilon(v) \qquad \text{in } L^2(\Omega;\mathbb{S}^d).$$

Since norms are weakly lower semicontinuous by Lemma 3.2, we obtain

$$a(v,v) = \|\varepsilon(v)\|_{\mathcal{A}}^2 \leq \liminf_{j\to\infty} \|\varepsilon(v_j)\|_{\mathcal{A}}^2 = \liminf_{j\to\infty} a(v_j,v_j).$$

Furthermore, $\ell \in V'$ is weakly continuous, hence $\ell(v_j) \to \ell(v)$. Therefore,

$$\mathcal{J}(v) = \frac{1}{2}\,a(v,v) - \ell(v) \leq \frac{1}{2}\,\liminf_{j\to\infty} a(v_j,v_j) - \lim_{j\to\infty} \ell(v_j) = \liminf_{j\to\infty} \mathcal{J}(v_j).$$

■

Proposition 3.2 (Existence of Minimizer of $\mathcal{J}$)**.** *The functional $\mathcal{J}$ defined in (3.1) has at least one minimizer $u \in K$.*

Proof. Set

$$M := \inf_{v\in K} \mathcal{J}(v) \in [-\infty,\infty).$$

Since $0 \in K$ we have $M \leq \mathcal{J}(0) = 0$, hence M is finite from above. Let $\{v_j\}_{j\in\mathbb{N}} \subset K$ be a minimizing sequence, i.e. $\mathcal{J}(v_j) \to M$ as $j \to \infty$. By Lemma 3.3 *i)*, $\mathcal{J}$ is coercive and thus, the sequence $\{v_j\}_{j\in\mathbb{N}}$ is bounded in V. Since V is a Hilbert space, there exists a subsequence $\{v_{j_k}\}_{k\in\mathbb{N}}$ and an element $u \in V$ such that

$$v_{j_k} \rightharpoonup u \qquad \text{in } V.$$

As $K \subset V$ is closed and convex, it is weakly closed in V. Hence $u \in K$. By Lemma 3.3 *ii)*, $\mathcal{J}$ is weakly lower semicontinuous and thus we obtain

$$\mathcal{J}(u) \leq \liminf_{k\to\infty} \mathcal{J}(v_{j_k}) = M = \inf_{v\in K} \mathcal{J}(v),$$

so u is a minimizer of $\mathcal{J}$ on K. ■

We collect the previous results in the following corollary and further give a standard stability estimate for the norm of the unique solution.

Corollary 3.1 (Well-posedness of Problem 3.1)**.** *The functional $\mathcal{J}$ defined in (3.1) admits a unique minimizer $u \in K$. Moreover, the minimizer satisfies the a-priori bound*

$$\|u\|_{1,\Omega} \leq \frac{2\,C_K^2\,\max\{1,C_{\mathrm{tr}}\}}{\alpha_{\mathcal{A}}}\,\left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right). \tag{3.3}$$

Consequently, Problem 3.1 has a unique stable solution.

Proof. By Proposition 3.2 we have existence of a solution $u \in K$ to Problem 3.1 and by Proposition 3.1 it is unique. It remains to prove the a-priori bound. Since u is the unique minimizer, it holds

$$\mathcal{J}(u) \leq \mathcal{J}(0) = 0 \qquad \Longrightarrow \qquad \frac{1}{2}\,a(u,u) \leq \ell(u).$$

By Proposition 2.4 *ii)* we have $a(u,u) \geq \alpha_{\mathcal{A}}/C_K^2\,\|u\|_{1,\Omega}^2$ and thus

$$\frac{\alpha_{\mathcal{A}}}{2\,C_K^2}\,\|u\|_{1,\Omega}^2 \leq \frac{1}{2}\,a(u,u) \leq |\ell(u)| \leq \max\{1,C_{\mathrm{tr}}\}\,\left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right)\|u\|_{1,\Omega},$$

where the last inequality is (2.5). Dividing both sides by $\|u\|_{1,\Omega}$ with $u \neq 0$ yields the desired estimate (3.3). For $u = 0$ the estimate is trivial. ■

4 Weak Formulations

Finally, we turn to the *weak formulation* of Signorini's problem. Unlike the weak formulation of conventional simple boundary value problems, here we do not end up with a variational equation, but with a variational inequality. This is derived by a first-order optimality condition on the functional $\mathcal{J}$ under the constraints of K, i.e. a condition on the *Fréchet derivative* of $\mathcal{J}$. To fix notation, we start with a definition.

Definition 4.1 (Fréchet derivative). *Let X and Y be normed vector spaces and let*

$$f : U \subseteq X \to Y$$

be a function defined on an open subset U of X. Then f is called `Fréchet differentiable` *at $x \in U$ if there exists a bounded linear operator $A(x) : X \to Y$ such that*

$$\lim_{\|h\|_X \to 0} \frac{\|f(x+h) - f(x) - A(x)[h]\|_Y}{\|h\|_X} = 0.$$

If $A(x)$ exists, it is unique and is called the `Fréchet derivative` *of f at x. We write*

$$f'(x)[h] = A(x)[h] \qquad \forall\, h \in X.$$

Finally, we prove Fréchet differentiability of the energy functional

$$\mathcal{J}(v) := \frac{1}{2}\, a(v,v) - \ell(v), \qquad v \in V$$

and provide an explicit formula for its Fréchet derivative.

Lemma 4.1 (Fréchet differentiability of $\mathcal{J}$). *The functional $\mathcal{J}$ defined in (3.1) is Fréchet differentiable on V. Its derivative at $u \in V$ is given by*

$$\mathcal{J}'(u)[h] = a(u,h) - \ell(h) \qquad \forall\, h \in V.$$

Proof. Let $u, k \in V$. Using bilinearity of a and linearity of ℓ, we obtain

$$\begin{aligned}
\mathcal{J}(u+k) - \mathcal{J}(u) &= \frac{1}{2}\big(a(u+k,u+k) - a(u,u)\big) - \ell(k) \\
&= \frac{1}{2}\big(a(u,u) + 2a(u,k) + a(k,k) - a(u,u)\big) - \ell(k) \\
&= a(u,k) - \ell(k) + \frac{1}{2}\, a(k,k).
\end{aligned}$$

Hence

$$\mathcal{J}(u+k) - \mathcal{J}(u) - \big(a(u,k) - \ell(k)\big) = \frac{1}{2}\, a(k,k).$$

Define $A(u) : V \to \mathbb{R}$ by

$$A(u)[h] := a(u,h) - \ell(h).$$

Since a and ℓ are bounded, $A(u)$ is linear and bounded. Moreover,

$$\frac{\big|\mathcal{J}(u+k) - \mathcal{J}(u) - A(u)[k]\big|}{\|k\|_{1,\Omega}} = \frac{1}{2}\,\frac{|a(k,k)|}{\|k\|_{1,\Omega}} \le \frac{1}{2}\, C_{\mathcal{A}}\, \|k\|_{1,\Omega} \xrightarrow{\|k\|_{1,\Omega} \to 0} 0,$$

by boundedness of a, (see Proposition 2.4). Therefore, $\mathcal{J}$ is Fréchet differentiable at u with

$$\mathcal{J}'(u)[h] = A(u)[h] = a(u,h) - \ell(h) \qquad \forall\, h \in V.$$

∎

4.1 Variational Inequality as an Optimality Condition

Proposition 4.1 (First-order optimality condition on convex sets). *A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality*

$$\mathcal{J}'(u)[v-u] \geq 0 \qquad \forall\, v \in K. \tag{4.1}$$

Proof. Assume first that $u \in K$ minimizes $\mathcal{J}$ over K and let $v \in K$. By convexity of K we have $u + t(v-u) \in K$ for all $t \in [0,1]$. Define $\phi(t) := \mathcal{J}(u + t(v-u))$. Then ϕ attains its minimum at $t = 0$, hence $\phi'(0^+) \geq 0$, where $\phi'(0^+)$ denotes the right-sided derivative at $t = 0$. Using the chain rule and Fréchet differentiability of $\mathcal{J}$ we obtain

$$\phi'(0^+) = \mathcal{J}'(u)[v-u] \geq 0.$$

Conversely, assume (4.1). Let $v \in K$ be arbitrary and set $h := v - u$. By Taylor expansion of the quadratic functional $\mathcal{J}$ we have

$$\mathcal{J}(u+h) - \mathcal{J}(u) = \mathcal{J}'(u)[h] + \frac{1}{2}\, a(h,h).$$

By (4.1), $\mathcal{J}'(u)[h] \geq 0$, and since $a(h,h) \geq 0$ we conclude $\mathcal{J}(v) \geq \mathcal{J}(u)$ for all $v \in K$. Hence u is a minimizer of $\mathcal{J}$ over K. ∎

Combining Lemma 4.1 and Proposition 4.1, $u \in K$ solves Problem 3.1 if and only if

$$\mathcal{J}'(u)[v-u] = a(u, v-u) - \ell(v-u) \geq 0 \qquad \forall\, v \in K.$$

This is the first weak formulation of Signorini's problem as a variational inequality.

Problem 4.1 (Signorini's Problem – Weak Formulation 1).

$$\text{Find } u \in K \text{ such that:} \qquad a(u, v-u) \geq \ell(v-u) \qquad \forall\, v \in K.$$

Well-posedness of Problem 4.1 is already ensured, since it is equivalent to the minimization Problem 3.1 by Proposition 4.1 in the sense that each solution of one of the problems is also a solution of the other. The solution is unique due to Corollary 3.1 with the stability estimate (3.3). Another equivalent formulation is the following.

Problem 4.2 (Signorini's Problem – Weak Formulation 2).

$$\text{Find } u \in K \text{ such that:} \qquad \begin{cases} a(u,u) = \ell(u), \\ a(u,v) \geq \ell(v) \qquad \forall\, v \in K. \end{cases}$$

Proposition 4.2 (Equivalence of Problem 4.1 and Problem 4.2). *Any solution $u \in K$ to Problem 4.1 is also a solution to Problem 4.2 and vice versa.*

Proof. Let $u \in K$ solve Problem 4.2. Then

$$a(u, v-u) = a(u,v) - a(u,u) = a(u,v) - \ell(u) \geq \ell(v) - \ell(u) = \ell(v-u) \qquad \forall\, v \in K.$$

Assuming that $u \in K$ solves Problem 4.1 and testing $v = 0 \in K$ yields

$$a(u,-u) \geq \ell(-u) \qquad \Longleftrightarrow \qquad -a(u,u) \geq -\ell(u),$$

which directly implies

$$a(u,u) \leq \ell(u).$$

However, testing with $v = 2u \in K$ yields $a(u,u) \geq \ell(u)$ and thus $a(u,u) = \ell(u)$. Furthermore, Problem 4.1 reads

$$a(u, v-u) = a(u,v) - a(u,u) \geq \ell(v-u) = \ell(v) - \ell(u) \qquad \forall\, v \in K,$$

which directly implies

$$a(u,v) \geq a(u,u) + \ell(v) - \ell(u) \qquad \forall\, v \in K.$$

Plugging in $a(u,u) = \ell(u)$ yields $a(u,v) \geq \ell(v)$ for all $v \in K$. ∎

4.2 Lipschitz Dependence of the Displacement on the Forces

We close this section by providing another stability result on the dependence of solutions to varying source terms as input data. This result is very similar to the stability estimate provided in Corollary 3.1. Its proof, however, is much more convenient at this point, where the weak formulations are available.

Proposition 4.3 (Lipschitz dependence on the data)**.** *For $i \in \{1,2\}$, let $f_i \in L^2(\Omega;\mathbb{R}^d)$ and $F_i \in L^2(\Gamma_N;\mathbb{R}^d)$, and let $\ell_i \in V'$ be the corresponding load functional, i.e.*

$$\ell_i(v) := \int_\Omega f_i \cdot v \,\mathrm{d}x + \int_{\Gamma_N} F_i \cdot \gamma v \,\mathrm{d}s, \qquad v \in V.$$

Moreover, let $u_i \in K$ be the unique solution of Problem 4.1 associated with ℓ_i, i.e.

$$a(u_i, v - u_i) \geq \ell_i(v - u_i) \qquad \forall\, v \in K.$$

Then the solution depends Lipschitz continuously on the load functional, i.e.

$$\|u_1 - u_2\|_{1,\Omega} \leq \frac{C_K^2}{\alpha_{\mathcal{A}}} \|\ell_1 - \ell_2\|_{V'}. \tag{4.2}$$

In particular, it depends Lipschitz continuously on the data (f_1, F_1) and (f_2, F_2) in the sense that

$$\|u_1 - u_2\|_{1,\Omega} \leq \frac{C_K^2 \max\{1, C_{\mathrm{tr}}\}}{\alpha_{\mathcal{A}}} \Big(\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N} \Big). \tag{4.3}$$

Proof. For u_1, choose $v = u_2 \in K$ in Problem 4.1. Then $a(u_1, u_2 - u_1) \geq \ell_1(u_2 - u_1)$. Similarly, choosing $v = u_1 \in K$ for u_2 yields $a(u_2, u_1 - u_2) \geq \ell_2(u_1 - u_2)$. Adding both inequalities and using symmetry of a, we obtain

$$a(u_1 - u_2, u_1 - u_2) \leq (\ell_1 - \ell_2)(u_1 - u_2).$$

By Proposition 2.4, the bilinear form a is V-elliptic with constant $\alpha_{\mathcal{A}}/C_K^2$. Hence

$$\frac{\alpha_{\mathcal{A}}}{C_K^2} \|u_1 - u_2\|_{1,\Omega}^2 \leq a(u_1 - u_2, u_1 - u_2) \leq (\ell_1 - \ell_2)(u_1 - u_2) \leq \|\ell_1 - \ell_2\|_{V'} \|u_1 - u_2\|_{1,\Omega}.$$

If $u_1 = u_2$, there is nothing to prove. Otherwise, dividing by $\|u_1 - u_2\|_{1,\Omega}$ yields (4.2). Moreover,

$$(\ell_1 - \ell_2)(v) = \int_\Omega (f_1 - f_2) \cdot v \,\mathrm{d}x + \int_{\Gamma_N} (F_1 - F_2) \cdot \gamma v \,\mathrm{d}s, \qquad v \in V.$$

Therefore, Proposition 2.4 applied to the functional $\ell_1 - \ell_2$ implies

$$\|\ell_1 - \ell_2\|_{V'} \leq \max\{1, C_{\mathrm{tr}}\} \Big(\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N} \Big).$$

Combining this with (4.2), we obtain (4.3). ∎

5 Strong Formulations

In this section, we pass from the variational setting to stronger formulations of Signorini's problem. This is in many respects the central part of the report, since here the contact conditions become progressively more explicit. We first introduce suitable normal traces for stresses in $H(\mathrm{div};\Omega)$ and use them to derive a hybrid formulation. Based on this, we then recover the contact conditions in a KKT-type form, where nonpenetration, sign condition, and complementarity are clearly separated. Finally, under additional regularity assumptions, we arrive at a classical strong formulation in which these contact conditions attain their most transparent form.

5.1 Stress Regularity and Normal Traces on $H(\mathrm{div};\Omega)$

The aim of this subsection is to develop a concept for normal traces of stress tensors $\sigma(u)$, in order to then subject them to contact conditions. Previously, in (2.3), for $v \in H^1(\Omega;\mathbb{R}^d)$, we defined its classical normal trace v_n as the trace $\gamma v \in H^{1/2}(\Gamma;\mathbb{R}^d)$ times the normal $\mathrm{n} \in L^\infty(\Gamma;\mathbb{R}^d)$. Now, since $u \in H^1(\Omega;\mathbb{R}^d)$, the stress tensor $\sigma(u)$ only belongs to $L^2(\Omega;\mathbb{S}^d)$. Therefore, in general, a trace operator γ is not available for $\sigma(u)$ and further regularity assumptions have to be taken on $\sigma(u)$, to get a notion for its trace. The canonical idea is to require $\sigma(u) \in H^1(\Omega;\mathbb{S}^d)$. In that case, $\gamma(\sigma(u)) \in H^{1/2}(\Gamma;\mathbb{S}^d)$ which makes it possible to define a classical normal trace $\gamma(\sigma(u))\mathrm{n} \in H^{1/2}(\Gamma;\mathbb{R}^d)$. As σ is an operator

$$\sigma : H^s(\Omega;\mathbb{R}^d) \to H^{s-1}(\Omega;\mathbb{S}^d), \qquad \sigma(u) = \mathcal{A} : \varepsilon(u),$$

this assumption further requires $u \in H^2(\Omega;\mathbb{R}^d)$ and also higher regularity on $\mathcal{A}$, which is much stronger than needed but will also be assumed later on in Section 5.5. A more relaxed assumption for now is $\sigma(u) \in H(\mathrm{div};\Omega)$, where

$$H(\mathrm{div};\Omega) := \left\{ \tau \in L^2(\Omega;\mathbb{S}^d) \;\middle|\; \mathrm{div}\,\tau \in L^2(\Omega;\mathbb{R}^d) \right\}$$

and to derive a concept for normal traces on $H(\mathrm{div};\Omega)$. Additionally to this assumption being a lot weaker, than $u \in H^2(\Omega;\mathbb{R}^d)$, it provides us with $\mathrm{div}\,\sigma(u) \in L^2(\Omega;\mathbb{R}^d)$, which is desired in strong formulations as strong bulk equilibrium equations $-\mathrm{div}\,\sigma(u) = f$ for source terms $f \in L^2(\Omega;\mathbb{R}^d)$ are now well-defined. It remains to provide a concept for normal traces on $H(\mathrm{div};\Omega)$. For sufficiently smooth $\tau : \overline{\Omega} \to \mathbb{S}^d$ and all sufficiently smooth $v : \overline{\Omega} \to \mathbb{R}^d$, *Green's identity* reads

$$\int_\Gamma (\gamma\tau)\,\mathrm{n} \cdot (\gamma v)\;\mathrm{d}s = \int_\Omega \tau : \nabla v\;\mathrm{d}x + \int_\Omega (\mathrm{div}\,\tau) \cdot v\;\mathrm{d}x \tag{5.1}$$

and can be interpreted as a definition of the normal trace $(\gamma\tau)\,\mathrm{n}$ in a weak sense, i.e. by integration against test functions. The key point now is, that the right-hand side of (5.1) is still well-defined if τ and v are not sufficiently smooth for the left-hand side to be well-defined, such as $\tau \in H(\mathrm{div};\Omega)$ and $v \in H^1(\Omega;\mathbb{R}^d)$, by understanding $\mathrm{div}\,\tau$ and ∇v in the weak sense. Hence, for fixed $\tau \in H(\mathrm{div};\Omega)$, we define

$$F_\tau(v) := \int_\Omega \tau : \nabla v\;\mathrm{d}x + \int_\Omega (\mathrm{div}\,\tau) \cdot v\;\mathrm{d}x, \qquad v \in H^1(\Omega;\mathbb{R}^d). \tag{5.2}$$

By Cauchy–Schwarz,

$$|F_\tau(v)| \leq (\|\tau\|_{0,\Omega} + \|\mathrm{div}\,\tau\|_{0,\Omega})\;\|v\|_{1,\Omega} \qquad \forall\, v \in H^1(\Omega;\mathbb{R}^d),$$

i.e. F_τ is a continuous linear functional on $H^1(\Omega;\mathbb{R}^d)$. Crucially, F_τ only depends on the trace $\gamma v \in H^{1/2}(\Gamma;\mathbb{R}^d)$ instead of the whole $v \in H^1(\Omega;\mathbb{R}^d)$.

Proposition 5.1 (Dependence of F_τ on the Trace)**.** *Let $\tau \in H(\mathrm{div};\Omega)$. Then, the functional F_τ defined in (5.2) vanishes on $H_0^1(\Omega;\mathbb{R}^d)$, i.e.*

$$F_\tau(w) = 0 \qquad \forall\, w \in H_0^1(\Omega;\mathbb{R}^d).$$

Consequently, F_τ depends only on the trace γv in the sense that

$$\gamma v_1 = \gamma v_2 \qquad \Longrightarrow \qquad F_\tau(v_1) = F_\tau(v_2) \qquad \forall\, v_1, v_2 \in H^1(\Omega;\mathbb{R}^d).$$

Proof. Let $w \in H_0^1(\Omega;\mathbb{R}^d)$. By density, there exists a sequence $(w_m)_{m\in\mathbb{N}} \subset \mathcal{D}(\Omega;\mathbb{R}^d)$ such that

$$w_m \to w \qquad \text{in } H^1(\Omega;\mathbb{R}^d).$$

Since $\tau \in H(\mathrm{div};\Omega)$, we have for every $m \in \mathbb{N}$ that

$$\int_\Omega (\mathrm{div}\,\tau)\cdot w_m \,\mathrm{d}x = -\int_\Omega \tau : \nabla w_m \,\mathrm{d}x,$$

and therefore

$$F_\tau(w_m) = 0 \qquad \forall\, m \in \mathbb{N}.$$

Passing to the limit $m \to \infty$ and using the continuity of F_τ on $H^1(\Omega;\mathbb{R}^d)$ yields

$$F_\tau(w) = \lim_{m\to\infty} F_\tau(w_m) = 0.$$

Hence F_τ vanishes on $H_0^1(\Omega;\mathbb{R}^d)$. Now let $v_1, v_2 \in H^1(\Omega;\mathbb{R}^d)$ satisfy $\gamma v_1 = \gamma v_2$. Then

$$\gamma(v_1 - v_2) = 0.$$

By the characterization $\ker(\gamma) = H_0^1(\Omega;\mathbb{R}^d)$, we obtain $v_1 - v_2 \in H_0^1(\Omega;\mathbb{R}^d)$. Therefore,

$$F_\tau(v_1) - F_\tau(v_2) = F_\tau(v_1 - v_2) = 0,$$

which proves the claim. ∎

By Proposition 5.1, the functional F_τ depends only on the trace γv. Hence the map

$$\gamma v \mapsto F_\tau(v)$$

defines a well-defined continuous linear functional on $H^{1/2}(\Gamma;\mathbb{R}^d)$. The dual space of $H^{1/2}(\Gamma;\mathbb{R}^d)$ is denoted by $H^{-1/2}(\Gamma;\mathbb{R}^d)$ and the duality pairing by $\langle\cdot,\cdot\rangle_\Gamma$. By this definition, there exists a unique element of $H^{-1/2}(\Gamma;\mathbb{R}^d)$ representing the functional F_τ.

Definition 5.1 (Normal Trace on $H(\mathrm{div};\Omega)$)**.** *Let $\tau \in H(\mathrm{div};\Omega)$. The* `normal trace` *of τ is the unique element*

$$\gamma_{\mathrm{n}}(\tau) \in H^{-1/2}(\Gamma;\mathbb{R}^d)$$

such that

$$\langle \gamma_{\mathrm{n}}(\tau), \gamma v\rangle_\Gamma = \int_\Omega \tau : \nabla v \,\mathrm{d}x + \int_\Omega (\mathrm{div}\,\tau)\cdot v \,\mathrm{d}x \qquad \forall\, v \in H^1(\Omega;\mathbb{R}^d). \tag{5.3}$$

In the smooth case, (5.3) reduces to (5.1), hence $\gamma_{\mathrm{n}}(\tau)$ coincides with the classical traction $(\gamma\tau)\,\mathrm{n}$. Coming back to our setting, we want to apply the derived theory to the symmetric stress tensor $\sigma(u) \in H(\mathrm{div};\Omega)$.

Proposition 5.2 (Symmetric tensors act only on the symmetric gradient)**.** *Let $\tau \in H(\mathrm{div};\Omega)$ and $v \in H^1(\Omega;\mathbb{R}^d)$. Then*

$$\tau : \nabla v = \tau : \varepsilon(v) \qquad \textit{a.e. in } \Omega.$$

Proof. Since $\tau \in H(\mathrm{div};\Omega) \subset L^2(\Omega;\mathbb{S}^d)$, the tensor τ is symmetric a.e. in Ω. Hence

$$\tau : \nabla v = \tau : \varepsilon(v) + \tau : (\nabla v - \varepsilon(v))\,.$$

The term

$$\nabla v - \varepsilon(v) = \frac{1}{2}(\nabla v - \nabla v^\top)$$

is skew-symmetric, whereas τ is symmetric. Therefore their Frobenius product vanishes a.e., i.e.

$$\tau : (\nabla v - \varepsilon(v)) = 0 \qquad \text{a.e. in } \Omega.$$

This proves

$$\tau : \nabla v = \tau : \varepsilon(v) \qquad \text{a.e. in } \Omega.$$

■

By Proposition 5.2, the normal trace $\sigma(u)\mathrm{n} := \gamma_{\mathrm{n}}(\sigma(u)) \in H^{-1/2}(\Gamma;\mathbb{R}^d)$ of the symmetric stress tensor $\sigma(u) \in H(\mathrm{div};\Omega)$ is defined via

$$\langle \sigma(u)\mathrm{n},\, \gamma v\rangle_\Gamma = \int_\Omega \sigma(u) : \varepsilon(v)\;\mathrm{d}x + \int_\Omega \mathrm{div}\,\sigma(u)\cdot v\;\mathrm{d}x \qquad \forall\, v \in H^1(\Omega;\mathbb{R}^d),$$

according to (5.3).

5.2 A Hybrid Formulation

The weak formulation in Problem 4.1 encodes the equilibrium equation and the contact conditions only implicitly through the variational inequality. As discussed in the previous Section, under the additional stress regularity $\sigma(u) \in H(\mathrm{div};\Omega)$, these conditions can be separated into a bulk equation in Ω and boundary relations involving the traction $\sigma(u)\mathrm{n}$. This leads to the following hybrid formulation, which may be viewed as an intermediate form between the weak variational inequality and the strong Signorini conditions in the later subsections.

Problem 5.1 (Signorini's Problem – Hybrid Formulation)**.** *For given $f \in L^2(\Omega;\mathbb{R}^d)$ and $F \in L^2(\Gamma_N;\mathbb{R}^d)$, find $u \in K$ such that*

$$\begin{cases} -\,\mathrm{div}\,\sigma(u) = f & \textit{a.e. in } \Omega, & (5.4a)\\ \langle \sigma(u)\mathrm{n}, \gamma u\rangle_\Gamma = \displaystyle\int_{\Gamma_N} F\cdot\gamma u\;\mathrm{d}s, & & (5.4b)\\ \langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma \geq \displaystyle\int_{\Gamma_N} F\cdot\gamma v\;\mathrm{d}s & \forall\, v \in K. & (5.4c)\end{cases}$$

We close this section by showing, that the hybrid formulation above is equivalent to the previous weak formulations in Problem 4.1 in order to prove that it is indeed a valid formulation of the Signorini problem. For practical reasons in the proof, we start with the following identity.

Lemma 5.1. *Let $f \in L^2(\Omega;\mathbb{R}^d)$. If the bulk equilibrium equation $-\,\mathrm{div}\,\sigma(u) = f$ holds almost everywhere in Ω, then*

$$a(u,v) - \ell(v) = \langle \sigma(u)\mathrm{n},\, \gamma v\rangle_\Gamma - \int_{\Gamma_N} F\cdot(\gamma v)\;\mathrm{d}s \qquad \forall\, v \in H^1(\Omega;\mathbb{R}^d). \tag{5.5}$$

Proof. Using the bulk equation and the definition of $a(u,v)$, we have

$$\langle \sigma(u)\mathrm{n},\, \gamma v\rangle_\Gamma = \int_\Omega \sigma(u) : \varepsilon(v)\;\mathrm{d}x + \int_\Omega \operatorname{div}\sigma(u)\cdot v\;\mathrm{d}x = a(u,v) - \int_\Omega f\cdot v\;\mathrm{d}x \qquad \forall\, v\in H^1(\Omega;\mathbb{R}^d).$$

By the definition of ℓ, this is

$$a(u,v) - \int_\Omega f\cdot v\; dx = a(u,v) - \ell(v) + \int_{\Gamma_N} F\cdot(\gamma v)\;\mathrm{d}s \qquad \forall\, v\in H^1(\Omega;\mathbb{R}^d),$$

which proves the claim. ∎

Proposition 5.3 (Equivalence of hybrid and weak formulation)**.** *If $u\in K$ solves Problem 5.1, then it also solves Problem 4.1. Conversely, if $u\in K$ solves Problem 4.1 and if, in addition, $\sigma(u)\in H(\operatorname{div};\Omega)$, then it also solves Problem 5.1.*

Proof. Let $u\in K$ solve the Hybrid Problem 5.1. Subtracting (5.4b) from (5.4c), yields

$$\langle \sigma(u)\mathrm{n},\, \gamma(v-u)\rangle_\Gamma - \int_{\Gamma_N} F\cdot\gamma(v-u)\;\mathrm{d}s \geq 0 \qquad \forall\, v\in K.$$

By (5.4a), we can directly apply the identity (5.5) with $v-u$ instead of v to obtain

$$a(u, v-u) - \ell(v-u) \geq 0 \qquad \forall\, v\in K,$$

which is exactly the weak formulation 4.1. Let now $u\in K$ solve the weak problem 4.1 with the additional assumption that $\sigma(u)\in H(\operatorname{div};\Omega)$. Let $\varphi\in\mathcal{D}(\Omega;\mathbb{R}^d)$ we a vector-valued test function on Ω. Since φ has compact support in Ω and thus a vanishing trace, $\gamma\varphi = 0$, we can perturb $u\in K$ by $\pm t\varphi, t\in\mathbb{R}$ without leaving K, i.e. $u\pm t\varphi\in K$. Testing both, $v = u+t\varphi$ and $v = u - t\varphi$, in the variational inequality of the weak formulation yields

$$a(u,t\varphi)\geq \ell(t\varphi) \qquad \text{and} \qquad a(u,-t\varphi)\geq\ell(-t\varphi).$$

By linearity of a in its second argument and linearity of ℓ, we can pull out t and $-t$ to obtain

$$a(u,\varphi)\geq\ell(\varphi) \qquad \text{and} \qquad a(u,\varphi)\leq\ell(\varphi)$$

and thus

$$a(u,\varphi) = \ell(\varphi) \qquad \forall\,\varphi\in\mathcal{D}(\Omega;\mathbb{R}^d). \tag{5.6}$$

The right-hand side of (5.6) is

$$\ell(\varphi) = \int_\Omega f\cdot\varphi\;\mathrm{d}x + \int_{\Gamma_N} F\cdot(\gamma\varphi)\;\mathrm{d}s = \int_\Omega f\cdot\varphi\;\mathrm{d}x,$$

where the boundary term vanished due to the compact support of φ. Furthermore, since $f\in L^2(\Omega;\mathbb{R}^d)$, it may be interpreted as a regular distribution, i.e.

$$\int_\Omega f\cdot\varphi\;\mathrm{d}x = \langle f,\,\varphi\rangle, \qquad \forall\,\varphi\in\mathcal{D}(\Omega;\mathbb{R}^d)$$

where $\langle\cdot,\,\cdot\rangle$ denotes the duality pairing between vector-valued distributions and vector-valued test functions. The right-hand side of (5.6) is by Proposition 5.2

$$a(u,\varphi) = \int_\Omega \sigma(u):\varepsilon(\varphi)\;\mathrm{d}x = \int_\Omega \sigma(u):\nabla\varphi\;\mathrm{d}x.$$

Since $\sigma(u)\in H(\operatorname{div};\Omega)\subset L^2(\Omega;\mathbb{S}^d)$ it may also be seen as a regular distribution and the weak divergence is exactly defined by

$$\langle -\operatorname{div}\sigma(u),\,\varphi\rangle = \int_\Omega \sigma(u):\nabla\varphi\;\mathrm{d}x \qquad \forall\,\varphi\in\mathcal{D}(\Omega;\mathbb{R}^d).$$

Thus, by (5.6), we have

$$\langle -\operatorname{div}\sigma(u),\, \varphi\rangle = \langle f,\, \varphi\rangle \qquad \forall\, \varphi \in \mathcal{D}(\Omega;\mathbb{R}^d). \tag{5.7}$$

Now since $\sigma(u) \in H(\operatorname{div};\Omega)$, we also have $\operatorname{div}\sigma(u) \in L^2(\Omega;\mathbb{R}^d)$ which means that (5.7) reads

$$-\operatorname{div}\sigma(u) = f \qquad \text{a.e. in } \Omega,$$

which is exactly the equilibrium bulk equation (5.4a). Next, we prove the hybrid inequality (5.4b). Since K is a cone, $0 \in K$ and $2u \in K$. Testing both, $v = 0$ and $v = 2u$, in the variational inequality of the weak formulation yields

$$a(u,-u) \geq \ell(-u) \qquad \text{and} \qquad a(u,u) \geq \ell(u)$$

and thus $a(u,u) = \ell(u)$. Using the identity (5.5) with u instead of v, we have

$$0 = a(u,u) - \ell(u) = \langle \sigma(u)\mathrm{n},\, \gamma u\rangle_\Gamma - \int_{\Gamma_N} F \cdot \gamma u \,\mathrm{d}s,$$

which directly yields the hybrid equality (5.4b). Moreover, with the identity (5.5) and the variational inequality from the weak formulation, we have

$$\langle \sigma(u)\mathrm{n},\, \gamma(v-u)\rangle_\Gamma - \int_{\Gamma_N} F \cdot \gamma(v-u)\,\mathrm{d}s = a(u,v-u) - \ell(v-u) \geq 0 \qquad \forall\, v \in K.$$

By linearity, this is

$$\langle \sigma(u)\mathrm{n},\, \gamma v\rangle_\Gamma \geq \int_{\Gamma_N} F \cdot \gamma v\,\mathrm{d}s + \left[\langle \sigma(u)\mathrm{n},\, \gamma u\rangle_\Gamma - \int_{\Gamma_N} F \cdot \gamma u\,\mathrm{d}s\right] \qquad \forall\, v \in V,$$

where the terms in the brackets cancel by the hybrid equality (5.4b). Therefore, also the hybrid inequality (5.4c) holds and the proof is completed. ■

5.3 Normal/Tangential Splitting and Local Normal Traces on $H(\operatorname{div};\Omega)$

In the hybrid formulation, the contact interaction is still encoded in the *global* normal trace

$$\sigma(u)\mathrm{n} = \gamma_{\mathrm{n}}(\sigma(u)) \in H^{-1/2}(\Gamma;\mathbb{R}^d),$$

which acts on traces on the whole boundary. We now restrict attention to the contact boundary Γ_C and introduce suitable contact trace spaces together with the corresponding local contact traction $\gamma_{\mathrm{n},C}$. Throughout this subsection and the next subsection, we assume $|\Gamma_N| = 0$. Hence

$$\Gamma = \overline{\Gamma_D} \cup \overline{\Gamma_C}, \qquad \Gamma_D \cap \Gamma_C = \emptyset$$

and every function in V has trace supported only on Γ_C. We define the so-called *Lions-Magenes* space

$$H^{1/2}_{00}(\Gamma_C;\mathbb{R}^d) := \left\{ g \in H^{1/2}(\Gamma_C;\mathbb{R}^d) \;\middle|\; \operatorname{Ext}_0(g) \in H^{1/2}(\Gamma;\mathbb{R}^d) \right\},$$

where $\operatorname{Ext}_0(g)$ denotes the zero extension of g from Γ_C to Γ. Since $|\Gamma_N| = 0$, the contact trace space is given by

$$W_C := \{\gamma v|_{\Gamma_C} \mid v \in V\} = H^{1/2}_{00}(\Gamma_C;\mathbb{R}^d).$$

Its norm is defined as the quotient norm

$$\|w\|_{W_C} := \inf\left\{\|v\|_{1,\Omega} \mid v \in V,\ \gamma v|_{\Gamma_C} = w\right\}, \qquad w \in W_C.$$

We now split contact traces into normal and tangential parts. Define

$$W_{C,\mathrm{n}} := \{w \cdot \mathrm{n} \mid w \in W_C\}, \qquad W_{C,t} := \{w \in W_C \mid w \cdot \mathrm{n} = 0\}.$$

Their quotient norms are given by

$$\|\eta\|_{W_{C,\mathrm{n}}} := \inf\{\|v\|_{1,\Omega} \mid v \in V,\ (\gamma v|_{\Gamma_C}) \cdot \mathrm{n} = \eta\}, \qquad \eta \in W_{C,\mathrm{n}},$$

and

$$\|\zeta\|_{W_{C,t}} := \inf\{\|v\|_{1,\Omega} \mid v \in V,\ (\gamma v|_{\Gamma_C})_t = \zeta\}, \qquad \zeta \in W_{C,t}.$$

Every $w \in W_C$ admits the decomposition $w = w_\mathrm{n}\mathrm{n} + w_t$, with $w_\mathrm{n} \in W_{C,\mathrm{n}}$ and $w_t \in W_{C,t}$. Hence

$$W_C = W_{C,\mathrm{n}}\mathrm{n} \oplus W_{C,t}.$$

We denote the dual spaces of W_C, $W_{C,\mathrm{n}}$, and $W_{C,t}$ by W_C', $W_{C,\mathrm{n}}'$ and $W_{C,t}'$, respectively, and use the generic notation $\langle\cdot,\,\cdot\rangle_{\Gamma_C}$ for the corresponding duality pairings.

Definition 5.2 (Contact normal trace)**.** *Let $\tau \in H(\mathrm{div};\Omega)$. The contact normal trace of τ on Γ_C is the functional*

$$\gamma_{\mathrm{n},C}(\tau) \in W_C'$$

defined by

$$\langle\gamma_{\mathrm{n},C}(\tau),\, g\rangle_{\Gamma_C} := \langle\gamma_\mathrm{n}(\tau),\, \mathrm{Ext}_0(g)\rangle_\Gamma \qquad \forall\, g \in W_C.$$

Since $W_C = H^{1/2}_{00}(\Gamma_C;\mathbb{R}^d)$ and $\mathrm{Ext}_0(g) \in H^{1/2}(\Gamma;\mathbb{R}^d)$ for every $g \in W_C$, this is well-defined. Moreover, if $g = \gamma v|_{\Gamma_C}$ for some $v \in V$, then $\gamma v = \mathrm{Ext}_0(g)$ on Γ and accordingly

$$\langle\gamma_{\mathrm{n},C}(\tau), g\rangle_{\Gamma_C} = \int_\Omega \tau : \nabla v \,\mathrm{d}x + \int_\Omega (\mathrm{div}\,\tau)\cdot v \,\mathrm{d}x.$$

In particular, the right-hand side depends only on the contact trace $g = \gamma v|_{\Gamma_C}$ and not on the chosen lift $v \in V$, similar as in Proposition 5.1.

Definition 5.3 (Normal and tangential components of contact tractions)**.** *Let $\lambda \in W_C'$. Its normal component $\lambda_{C,\mathrm{n}} \in W_{C,\mathrm{n}}'$ and tangential component $\lambda_{C,t} \in W_{C,t}'$ are defined by*

$$\langle\lambda_{C,\mathrm{n}},\, \eta\rangle_{\Gamma_C} := \langle\lambda,\, \eta\,\mathrm{n}\rangle_{\Gamma_C} \qquad \forall\, \eta \in W_{C,\mathrm{n}},$$

and

$$\langle\lambda_{C,t},\, \zeta\rangle_{\Gamma_C} := \langle\lambda,\, \zeta\rangle_{\Gamma_C} \qquad \forall\, \zeta \in W_{C,t}.$$

From now on, we denote by

$$\gamma_C v := (\gamma v)|_{\Gamma_C} \in W_C$$

the trace of $v \in V$ restricted to the contact boundary. Its normal and tangential components are denoted by

$$v_\mathrm{n} := \gamma_C v \cdot \mathrm{n} \in W_{C,\mathrm{n}}, \qquad v_t := \gamma_C v - v_\mathrm{n}\mathrm{n} \in W_{C,t}.$$

With a slight abuse of notation, we use the same symbols v_n and v_t for the restrictions to Γ_C of the global normal and tangential components of γv. Likewise, for $w \in W_C$ we write

$$w = w_\mathrm{n}\mathrm{n} + w_t, \qquad w_\mathrm{n} \in W_{C,\mathrm{n}}, \quad w_t \in W_{C,t}.$$

Moreover, we identify $W_{C,\mathrm{n}}'$ and $W_{C,t}'$ with the corresponding subspaces of W_C', so that for $\lambda \in W_C'$ and $w = w_\mathrm{n}\mathrm{n} + w_t \in W_C$,

$$\langle\lambda, w\rangle_{\Gamma_C} = \langle\lambda_{C,\mathrm{n}}, w_\mathrm{n}\rangle_{\Gamma_C} + \langle\lambda_{C,t}, w_t\rangle_{\Gamma_C}.$$

Applying this construction to the stress tensor, we define the contact traction functional by

$$\sigma_C(u)\mathrm{n} := \gamma_{\mathrm{n},C}(\sigma(u)) \in W_C'.$$

Its normal and tangential components are denoted by $\sigma_{C,\mathrm{n}}(u) \in W_{C,\mathrm{n}}'$ and $\sigma_{C,t}(u) \in W_{C,t}'$, respectively, such that

$$\sigma_C(u)\mathrm{n} = \sigma_{C,\mathrm{n}}(u)\mathrm{n} + \sigma_{C,t}(u) \in W_C'. \tag{5.8}$$

To encode the Signorini conditions, we introduce the cone of weakly nonpositive normal traces

$$W_{C,\mathrm{n}}^- := \{\eta \in W_{C,\mathrm{n}} \mid \eta \leq 0 \text{ a.e. on } \Gamma_C\},$$

and its dual cone of weakly nonpositive normal contact forces

$$\Lambda_C := \left\{\lambda \in W_{C,\mathrm{n}}' \;\middle|\; \langle \lambda,\, \eta\rangle_{\Gamma_C} \geq 0 \;\forall\, \eta \in W_{C,\mathrm{n}}^-\right\}.$$

If λ is represented by an L^2-function, then the condition $\lambda \in \Lambda_C$ is precisely the weak formulation of the condition $\lambda \leq 0$ almost everywhere on Γ_C, i.e. of weakly nonpositive normal contact forces.

5.4 KKT-Type Strong Formulation

We now reformulate the contact conditions in a form that separates the normal and tangential contact reactions on Γ_C. This leads to a *KKT-type formulation* of Signorini's problem, where the contact conditions are expressed through *primal feasibility*, *dual feasibility*, and *complementarity*. Here, "KKT" refers to the *Karush–Kuhn–Tucker* conditions from constrained optimization. In the present setting, the nonpenetration condition $u_\mathrm{n} \leq 0$ on Γ_C plays the role of the primal inequality constraint, the condition $\sigma_{C,\mathrm{n}}(u) \in \Lambda_C$ encodes the weak nonpositivity of the normal contact force, and the complementarity relation $\langle \sigma_{C,\mathrm{n}}(u),\, u_\mathrm{n}\rangle_{\Gamma_C} = 0$ expresses that contact pressure can occur only where the gap vanishes.

Problem 5.2 (Signorini's Problem – Strong Formulation with KKT Contact Conditions)**.** *For given $f \in L^2(\Omega;\mathbb{R}^d)$, find $u \in V$ such that*

$$\begin{cases} -\operatorname{div}\sigma(u) = f & \text{a.e. in } \Omega, \\ \sigma_{C,t}(u) = 0 & \text{in } W_{C,t}', \\ u_\mathrm{n} \leq 0 & \text{a.e. on } \Gamma_C, \\ \sigma_{C,\mathrm{n}}(u) \in \Lambda_C, & \\ \langle \sigma_{C,\mathrm{n}}(u),\, u_\mathrm{n}\rangle_{\Gamma_C} = 0. & \end{cases} \begin{matrix} (5.9\mathrm{a}) \\ (5.9\mathrm{b}) \\ (5.9\mathrm{c}) \\ (5.9\mathrm{d}) \\ (5.9\mathrm{e}) \end{matrix}$$

Under the standing assumption $|\Gamma_N| = 0$, this formulation is equivalent to the hybrid formulation introduced above. This equivalence will be proved now and will serve as the basis for the mixed formulation in Section 6.

Proposition 5.4 (Equivalence of KKT and Hybrid Formulation)**.** *Suppose that $|\Gamma_N| = 0$. Then every solution to Problem 5.2 is also a solution to Problem 5.1 and vice versa.*

Proof. Assume first that $u \in V$ solves the above KKT formulation 5.2 of Signorini's Problem. The bulk equation $-\operatorname{div}\sigma(u) = f$ almost everywhere in Ω is contained in both formulations and by (5.9c), $u \in K$. It remains to show the hybrid equality (5.4b) and inequality (5.4c), where in each the right-hand sides vanish, due to $|\Gamma_N| = 0$. Let $v \in K$ be arbitrary. Since $|\Gamma_N| = 0$ and $(\gamma v)|_{\Gamma_D} = 0$, we have

$$\langle \sigma(u)\mathrm{n},\, \gamma v\rangle_\Gamma = \langle \sigma_C(u)\mathrm{n},\, \gamma_C v\rangle_{\Gamma_C}.$$

Using the decomposition (5.8) of $\sigma_C(u)\mathrm{n}$ into normal and tangential components together with Definition 5.2, we get

$$\langle \sigma_C(u)\mathrm{n},\, \gamma v\rangle_{\Gamma_C} = \langle \sigma_{C,\mathrm{n}}(u),\, v_\mathrm{n}\rangle_{\Gamma_C} + \langle \sigma_{C,t}(u),\, v_t\rangle_{\Gamma_C}.$$

By (5.9b), the second term vanishes. Therefore, $\langle \sigma(u)\mathrm{n},\, \gamma v\rangle_\Gamma = \langle \sigma_{C,\mathrm{n}}(u),\, v_\mathrm{n}\rangle_{\Gamma_C}$. Since $v \in K$, we have $v_\mathrm{n} \le 0$ almost everywhere on Γ_C, and by (5.9d) we know that $\sigma_{C,\mathrm{n}}(u) \in \Lambda_C$. Hence

$$\langle \sigma_C(u)\mathrm{n},\, \gamma_C v\rangle_{\Gamma_C} = \langle \sigma_{C,\mathrm{n}}(u),\, v_\mathrm{n}\rangle_{\Gamma_C} \ge 0,$$

which proves the inequality (5.4c). Taking $v = u$ and using (5.9e), we similarly obtain

$$\langle \sigma(u)\mathrm{n},\, \gamma u\rangle_\Gamma = \langle \sigma_{C,\mathrm{n}}(u),\, u_\mathrm{n}\rangle_{\Gamma_C} = 0,$$

which is exactly the hybrid equality (5.4b). Thus u solves Problem 5.1. Conversely, assume that $u \in K$ solves Problem 5.1. Then $u \in V$, and (5.9c) is already contained in the definition of K. Moreover, (5.4a) is exactly (5.9a). We next prove (5.9b). Let $w_t \in W_{C,t}$. Then there exists $w \in V$ such that

$$\gamma w|_{\Gamma_C} = w_t \qquad \text{and} \qquad w_\mathrm{n} = 0 \text{ on } \Gamma_C.$$

Since $u \in K$ and $w_\mathrm{n} = 0$ on Γ_C, we have $u \pm w \in K$. Hence, the hybrid inequality (5.4c) applied to $v = u + w$ and $v = u - w$, together with the hybrid equality (5.4b), yields

$$\langle \sigma(u)\mathrm{n},\, \gamma w\rangle_\Gamma \ge 0 \qquad \text{and} \qquad \langle \sigma(u)\mathrm{n},\, \gamma w\rangle_\Gamma \le 0.$$

Therefore, $\langle \sigma(u)\mathrm{n},\, \gamma w\rangle_\Gamma = 0$. Again using $|\Gamma_N| = 0$ and $\gamma w = 0$ on Γ_D, we obtain

$$0 = \langle \sigma(u)\mathrm{n},\, \gamma w\rangle_\Gamma = \langle \sigma_C(u)\mathrm{n},\, \gamma_C w\rangle_{\Gamma_C} = \langle \sigma_{C,\mathrm{n}}(u),\, w_\mathrm{n}\rangle_{\Gamma_C} + \langle \sigma_{C,t}(u),\, w_t\rangle_{\Gamma_C} = \langle \sigma_{C,t}(u),\, w_t\rangle_{\Gamma_C},$$

since $w_\mathrm{n} = 0$ on Γ_C. Since $w_t \in W_{C,t}$ was arbitrary, this proves (5.9b). We now prove (5.9d). Let $\eta \in W^-_{C,\mathrm{n}}$, i.e. $\eta \le 0$ a.e. on Γ_C. Then there exists $v \in V$ such that $v_\mathrm{n} = \eta$. Because $\eta \le 0$, we have $v \in K$. Hence the hybrid inequality (5.4c) gives $\langle \sigma(u)\mathrm{n},\, \gamma v\rangle_\Gamma \ge 0$. Using $|\Gamma_N| = 0$, $\gamma v = 0$ on Γ_D, and (5.9b), we infer

$$0 \le \langle \sigma_C(u)\mathrm{n}, \gamma_C v\rangle_{\Gamma_C} = \langle \sigma_{C,\mathrm{n}}(u), v_\mathrm{n}\rangle_{\Gamma_C} + \langle \sigma_{C,t}(u), v_t\rangle_{\Gamma_C} = \langle \sigma_{C,\mathrm{n}}(u), \eta\rangle_{\Gamma_C},$$

which proves $\sigma_{C,\mathrm{n}}(u) \in \Lambda_C$, i.e. (5.9d). Finally, the hybrid equality (5.4b), together with $|\Gamma_N| = 0$, $\gamma u = 0$ on Γ_D, and (5.9b), yields

$$0 = \langle \sigma(u)\mathrm{n}, \gamma u\rangle_\Gamma = \langle \sigma_C(u)\mathrm{n}, \gamma_C u\rangle_{\Gamma_C} = \langle \sigma_{C,\mathrm{n}}(u), u_\mathrm{n}\rangle_{\Gamma_C} + \langle \sigma_{C,t}(u), u_t\rangle_{\Gamma_C} = \langle \sigma_{C,\mathrm{n}}(u), u_\mathrm{n}\rangle_{\Gamma_C},$$

which is exactly (5.9e). Hence u solves Problem 5.2. ■

5.5 Classical Strong Formulation under Additional Regularity

So far, the displacement field was sought only in $u \in V \subset H^1(\Omega;\mathbb{R}^d)$. Accordingly, the stress tensor satisfies only $\sigma(u) \in L^2(\Omega;\mathbb{S}^d)$ or, under the additional assumption discussed in Section 5.1, $\sigma(u) \in H(\mathrm{div};\Omega)$, so that no classical boundary trace of $\sigma(u)$ is available in general but only the weak concept of normal traces $\gamma_\mathrm{n}(\sigma(u)) \in H^{-1/2}(\Gamma;\mathbb{R}^d)$ on $H(\mathrm{div};\Omega)$. In order to formulate the boundary traction and the contact conditions pointwise on Γ_N and Γ_C, we now impose additional regularity on the displacement. We assume that

$$u \in H^s(\Omega;\mathbb{R}^d) \qquad \text{with} \qquad s > \frac{3}{2}.$$

According to Definition 2.6, the strain $\varepsilon(u)$ imposed by $u \in H^s(\Omega;\mathbb{R}^d)$ is now a function in $H^{s-1}(\Omega;\mathbb{S}^d)$. We further assume that the elasticity tensor field $\mathcal{A} : \Omega \to \mathcal{L}(\mathbb{S}^d;\mathbb{S}^d)$ is sufficiently regular so that

$$\sigma(u) = \mathcal{A} : \varepsilon(u) \in H^{s-1}(\Omega;\mathbb{S}^d).$$

For instance, this is the case if $\mathcal{A}$ is constant. More generally, for $\frac{3}{2} < s \leq 2$, it is sufficient to assume

$$\mathcal{A}_{ijkl} \in W^{1,\infty}(\Omega) \qquad \forall\, 1 \leq i,j,k,l \leq d.$$

In order to apply the trace theorem to $\sigma(u)$, we additionally assume

$$\Gamma \in \mathcal{C}^{m,1} \qquad \text{with} \qquad m \geq \lceil s-1 \rceil.$$

Then $\gamma(\sigma(u)) \in H^{s-3/2}(\Gamma;\mathbb{S}^d)$, and, since now the unit outward normal satisfies $\mathrm{n} \in \mathcal{C}^{m-1,1}(\Gamma;\mathbb{R}^d)$, the classical boundary traction

$$t(u) := (\gamma\sigma(u))\,\mathrm{n}$$

is well-defined and belongs to $H^{s-3/2}(\Gamma;\mathbb{R}^d)$. Its normal and tangential components are given by

$$\sigma_{\mathrm{n}}(u) := t(u)\cdot \mathrm{n} \in H^{s-3/2}(\Gamma), \qquad \sigma_t(u) := t(u) - \sigma_{\mathrm{n}}(u)\,\mathrm{n} \in H^{s-3/2}(\Gamma;\mathbb{R}^d).$$

Likewise, the displacement trace satisfies $\gamma u \in H^{s-1/2}(\Gamma;\mathbb{R}^d)$, with normal and tangential components

$$u_{\mathrm{n}} := (\gamma u)\cdot \mathrm{n} \in H^{s-1/2}(\Gamma), \qquad u_t := \gamma u - u_{\mathrm{n}}\,\mathrm{n} \in H^{s-1/2}(\Gamma;\mathbb{R}^d).$$

If, in addition, $\sigma(u) \in H(\mathrm{div};\Omega)$, then the normal trace on $H(\mathrm{div};\Omega)$ and the classical normal trace on $H^s(\Omega;\mathbb{S}^d)$ coincide in the sense that

$$\langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma = \int_\Gamma t(u)\cdot \gamma v \,\mathrm{d}s \qquad \forall\, v \in V. \tag{5.10}$$

This allows us to pass from the hybrid formulation to a classical strong formulation with pointwise contact conditions.

Problem 5.3 (Signorini's Problem – Classical Strong Formulation)**.** *For given* $f \in L^2(\Omega;\mathbb{R}^d)$ *and* $F \in L^2(\Gamma_N;\mathbb{R}^d)$ *find* $u \in H^s(\Omega;\mathbb{R}^d)$ *with* $s > 3/2$ *such that the* small strain elasticity equations

$$\begin{cases} -\operatorname{div}\sigma(u) = f & \text{in } \Omega, & (5.11\text{a})\\ u = 0 & \text{on } \Gamma_D, & (5.11\text{b})\\ t(u) = F & \text{on } \Gamma_N, & (5.11\text{c}) \end{cases}$$

and the contact conditions

$$\begin{cases} u_{\mathrm{n}} \leq 0 & \text{on } \Gamma_C, & (5.12\text{a})\\ \sigma_{\mathrm{n}}(u) \leq 0 & \text{on } \Gamma_C, & (5.12\text{b})\\ \sigma_{\mathrm{n}}(u)\,u_{\mathrm{n}} = 0 & \text{on } \Gamma_C, & (5.12\text{c})\\ \sigma_t(u) = 0 & \text{on } \Gamma_C & (5.12\text{d}) \end{cases}$$

hold.

The classical strong formulation admits a direct mechanical interpretation. The equilibrium equation (5.11a) expresses, just as previously, the balance of internal stresses and body forces in the bulk. The homogeneous Dirichlet condition (5.11b) on Γ_D means, just as before, that the body is clamped along Γ_D, whereas the Neumann condition (5.11c) prescribes the boundary traction acting on the free part of the boundary. On the potential contact boundary Γ_C, the condition $u_{\mathrm{n}} \leq 0$ is exactly the *nonpenetration* constraint, previously encoded in K, while $\sigma_{\mathrm{n}}(u) \leq 0$ states that the foundation can exert only *compressive normal forces*. The *complementarity relation* $\sigma_{\mathrm{n}}(u)\,u_{\mathrm{n}} = 0$ connects these two conditions. Either the body is detached from the foundation, in which case no normal contact pressure acts, or the body is in contact, in which case the normal gap vanishes. Finally, the *tangential condition* $\sigma_t(u) = 0$ describes frictionless contact, meaning that the foundation does not exert any tangential force on the body.

Proposition 5.5 (Equivalence of Classical Strong and Hybrid Formulation)**.** *Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$ solve Problem 5.3. Then, it also solves Problem 5.1. Conversely, if $u \in K$ solves Problem 5.1 and if, in addition, $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$, then it also solves Problem 5.3.*

Proof. Again, the bulk equation $-\operatorname{div} \sigma(u) = f$ almost everywhere in Ω appears in both formulations. Let $u \in H^s(\Omega; \mathbb{R}^d)$, $s > 3/2$ solve Problem 5.3. With (5.11b) and (5.12a) we have $u \in K$. It remains to prove the hybrid inequality (5.4c) and equality (5.4b), which here read

$$\int_\Gamma t(u) \cdot (\gamma v) \,\mathrm{d}s \geq \int_{\Gamma_N} F \cdot (\gamma v) \,\mathrm{d}s \qquad \forall\, v \in K \tag{5.13}$$

and

$$\int_\Gamma t(u) \cdot (\gamma u) \,\mathrm{d}s = \int_{\Gamma_N} F \cdot (\gamma u) \,\mathrm{d}s, \tag{5.14}$$

due to the sufficient regularity of u such that the normal trace $\sigma(u)\mathrm{n}$ and the classical traction $t(u)$ coincide in the sense of (5.10). We decompose

$$\int_\Gamma t(u) \cdot (\gamma v) \,\mathrm{d}s = \int_{\Gamma_D} t(u) \cdot (\gamma v) \,\mathrm{d}s + \int_{\Gamma_N} t(u) \cdot (\gamma v) \,\mathrm{d}s + \int_{\Gamma_C} t(u) \cdot (\gamma v) \,\mathrm{d}s. \tag{5.15}$$

By $(\gamma v)|_{\Gamma_D} = 0$ and the Neumann condition (5.11c), we are left over with

$$\int_\Gamma t(u) \cdot (\gamma v) \,\mathrm{d}s = \int_{\Gamma_N} F \cdot (\gamma v) \,\mathrm{d}s + \int_{\Gamma_C} t(u) \cdot (\gamma v) \,\mathrm{d}s.$$

With the splittings $t(u) = \sigma_\mathrm{n}(u)\mathrm{n} + \sigma_t(u)$ and $\gamma v = v_\mathrm{n}\mathrm{n} + v_t$ in their normal and tangential components, the integrand of the right term is, on Γ_C, given by

$$t(u) \cdot (\gamma v) = \sigma_\mathrm{n}(u)\, v_\mathrm{n} + \sigma_t(u) \cdot v_t = \sigma_\mathrm{n}(u)\, v_\mathrm{n},$$

where the tangential term vanishes on Γ_C due to (5.12d). Furthermore, since $v \in K$, we have $v_\mathrm{n} \leq 0$ on Γ_C. Together with (5.12b), it follows that $\sigma_\mathrm{n}(u) v_\mathrm{n} \geq 0$ on Γ_C, i.e.

$$\int_\Gamma t(u) \cdot (\gamma v) \,\mathrm{d}s = \int_{\Gamma_N} F \cdot (\gamma v) \,\mathrm{d}s + \int_{\Gamma_C} \sigma_\mathrm{n}(u)\, v_\mathrm{n} \,\mathrm{d}s \geq \int_{\Gamma_N} F \cdot (\gamma v) \,\mathrm{d}s \qquad \forall\, v \in K,$$

which is exactly the hybrid inequality (5.13). By the same arguments with $v = u$, we arrive at

$$\int_\Gamma t(u) \cdot (\gamma u) \,\mathrm{d}s = \int_{\Gamma_N} F \cdot (\gamma u) \,\mathrm{d}s + \int_{\Gamma_C} \sigma_\mathrm{n}(u)\, u_\mathrm{n} = \int_{\Gamma_N} F \cdot (\gamma u) \,\mathrm{d}s,$$

where the Γ_C-term vanishes due to the complementarity condition (5.12c). This proves the hybrid equality (5.14). Therefore, u is also a solution to Problem 5.1. Now, let $u \in K \cap H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$ solve Problem 5.1. The requirement $u \in K$ immediately implies (5.11b) and (5.12a). We now prove the Neumann condition $t(u) = F$ on Γ_N, i.e. (5.11c). For some $g \in H^{1/2}_{00}(\Gamma_N; \mathbb{R}^d)$ choose $v \in H^1(\Omega; \mathbb{R}^d)$ such that

$$(\gamma v)|_{\Gamma_N} = g, \qquad (\gamma v)|_{\Gamma_C} = 0, \qquad (\gamma v)|_{\Gamma_D} = 0.$$

Since $v_\mathrm{n} = 0$ on Γ_C and $\gamma v = 0$ on Γ_D, we have $v \in K$ and also $-v \in K$. Thus, we can test $\pm v$ in the hybrid inequality (5.13) to obtain

$$\int_{\Gamma_N} t(u) \cdot g \,\mathrm{d}s \geq \int_{\Gamma_N} F \cdot g \,\mathrm{d}s, \qquad \text{and} \qquad -\int_{\Gamma_N} t(u) \cdot g \,\mathrm{d}s \geq -\int_{\Gamma_N} F \cdot g \,\mathrm{d}s$$

for all $g \in H^{1/2}_{00}(\Gamma_N; \mathbb{R}^d)$. Therefore,

$$\int_{\Gamma_N} (t(u) - F) \cdot g \,\mathrm{d}s = 0 \qquad \forall\, g \in H^{1/2}_{00}(\Gamma_N; \mathbb{R}^d),$$

i.e. $t(u) = F$ on Γ_N which is exactly (5.11c) since both, $t(u)$ and F belong to $L^2(\Gamma_N;\mathbb{R}^d)$. We continue by showing (5.12d). Plugging (5.15) in to (5.13) and using $t(u) = F$ on Γ_N as well as $(\gamma v)|_{\Gamma_D} = 0$ for $v \in K$, we have

$$\int_{\Gamma_C} t(u) \cdot (\gamma v)\, \mathrm{d}s \geq 0 \qquad \forall\, v \in K. \tag{5.16}$$

Let $g_t \in H^{1/2}_{00}(\Gamma_C;\mathbb{R}^d)$ with $g_t \cdot \mathrm{n} = 0$, i.e. $g_t \in W_{C,t}$ and choose $v \in H^1(\Omega;\mathbb{R}^d)$ with

$$(\gamma v)|_{\Gamma_C} = g_t, \qquad (\gamma v)|_{\Gamma_N} = 0, \qquad (\gamma v)|_{\Gamma_D} = 0.$$

Again, $\pm v \in K$ and therefore

$$\int_{\Gamma_C} t(u) \cdot g_t\, \mathrm{d}s \geq 0, \qquad \text{and} \qquad -\int_{\Gamma_C} t(u) \cdot g_t\, \mathrm{d}s \geq 0.$$

Splitting $t(u) = \sigma_{\mathrm{n}}(u)\mathrm{n} + \sigma_t(u)$ into its normal and tangential components, this means

$$\int_{\Gamma_C} (\sigma_{\mathrm{n}}(u)\mathrm{n} + \sigma_t(u)) \cdot g_t\, \mathrm{d}s = \int_{\Gamma_C} \sigma_t(u) \cdot g_t\, \mathrm{d}s + \int_{\Gamma_C} \sigma_{\mathrm{n}}(u)(\mathrm{n} \cdot g_t)\, \mathrm{d}s = 0. \qquad \forall\, g_t \in W_{C,t}$$

With $g_t \cdot \mathrm{n} = 0$, we thus have

$$\int_{\Gamma_C} \sigma_t(u) \cdot g_t\, \mathrm{d}s = 0 \qquad \forall\, g_t \in W_{C,t}$$

which means $\sigma_t(u) = 0$ on Γ_C, i.e (5.12d). Next, we prove (5.12b). Let $\eta \in H^{1/2}_{00}(\Gamma_C)$ with $\eta \leq 0$ on Γ_C, i.e. $\eta \in W^-_{C,\mathrm{n}}$ and choose $v \in H^1(\Omega;\mathbb{R}^d)$ with

$$(\gamma v)|_{\Gamma_C} = \eta\, \mathrm{n}, \qquad (\gamma v)|_{\Gamma_N} = 0, \qquad (\gamma v)|_{\Gamma_D} = 0.$$

Then, $v_{\mathrm{n}} = (\gamma v) \cdot \mathrm{n} = \eta \leq 0$ on Γ_C, i.e. $v \in K$ and thus, plugging v into (5.16), we have

$$\int_{\Gamma_C} t(u) \cdot (\eta\, \mathrm{n})\, \mathrm{d}s \geq 0 \qquad \forall\, \eta \in W^-_{C,\mathrm{n}}.$$

Again, by splitting $t(u) = \sigma_{\mathrm{n}}(u)\mathrm{n} + \sigma_t(u)$ and using $\sigma_t(u) = 0$ on Γ_C, this yields

$$\int_{\Gamma_C} \sigma_{\mathrm{n}}(u)\, \eta\, \mathrm{d}s \geq 0 \qquad \forall\, \eta \in W^-_{C,\mathrm{n}}$$

or, in other words,

$$\int_{\Gamma_C} \sigma_{\mathrm{n}}(u)\, \eta\, \mathrm{d}s \leq 0 \qquad \forall\, \varphi \in H^{1/2}_{00}(\Gamma_C),\ \varphi \geq 0,$$

which implies $\sigma_{\mathrm{n}}(u) \leq 0$ on Γ_C, i.e. (5.12b). It remains to show the complementarity condition (5.12c). Plugging (5.15) for $v = u$ into (5.14) and using $t(u) = F$ on Γ_N as well as the Dirichlet condition $(\gamma u)|_{\Gamma_D} = 0$, yields

$$\int_{\Gamma_C} t(u) \cdot (\gamma u)\, \mathrm{d}s = \int_{\Gamma_C} \sigma_{\mathrm{n}}(u)\, u_{\mathrm{n}}\, \mathrm{d}s = 0,$$

with $t(u) = \sigma_{\mathrm{n}}(u)\mathrm{n}$ as previously. The integrand $\sigma_{\mathrm{n}}(u)\, u_{\mathrm{n}}$ is nonnegative on Γ_C since $u_{\mathrm{n}} \leq 0$ on Γ_C and $\sigma_{\mathrm{n}}(u) \leq 0$ on Γ_C. Therefore $\sigma_{\mathrm{n}}(u)\, u_{\mathrm{n}} = 0$ on Γ_C which is exactly (5.12c). This completes the proof. ■

6 Mixed Formulation

The KKT formulation of the previous section already separates the normal contact reaction from the displacement field. This suggests to introduce the normal contact force as an additional unknown. In other words, instead of recovering the normal contact stress $\sigma_{C,\mathrm{n}}(u)$ a posteriori from u, we now treat it as an independent variable

$$\lambda \in \Lambda_C \subset W'_{C,\mathrm{n}}.$$

This leads to a mixed weak formulation of Signorini's problem. The role of λ is that of a Lagrange multiplier associated with the nonpenetration constraint $u_\mathrm{n} \leq 0$ on Γ_C. In the present setting, it represents the normal contact force on the rigid foundation. This point of view is consistent with the duality principle developed in [1, Section 7.1], where mixed formulations are obtained by introducing a multiplier for the unilateral contact constraint.

Problem 6.1 (Signorini's Problem – Mixed Weak Formulation)**.** *For given $f \in L^2(\Omega; \mathbb{R}^d)$ find $(u, \lambda) \in V \times \Lambda_C$ such that*

$$\begin{cases} a(u,v) - \langle \lambda,\, v_\mathrm{n} \rangle_{\Gamma_C} = \ell(v) & \forall\, v \in V, \qquad (6.1a) \\ \langle \mu - \lambda,\, u_\mathrm{n} \rangle_{\Gamma_C} \geq 0 & \forall\, \mu \in \Lambda_C. \qquad (6.1b) \end{cases}$$

The first equation expresses the weak balance of forces, where the contact pressure is now represented explicitly by the multiplier λ. The second inequality is the weak counterpart of the unilateral contact conditions. It enforces compatibility between the admissible normal displacement u_n and the weakly nonpositive contact force $\lambda \in \Lambda_C$.

Proposition 6.1 (Equivalence of mixed and KKT formulation)**.** *Assume $|\Gamma_N| = 0$. If u solves Problem 5.2, then the pair $(u, \sigma_{C,\mathrm{n}}(u)) \in V \times \Lambda_C$ solves the mixed Problem 6.1. Conversely, if $(u, \lambda) \in V \times \Lambda_C$ solves the mixed Problem 6.1, then u solves Problem 5.2 and $\lambda = \sigma_{C,\mathrm{n}}(u)$.*

Proof. Assume first that u solves Problem 5.2. Define $\lambda := \sigma_{C,\mathrm{n}}(u) \in \Lambda_C$, where the inclusion in Λ_C is exactly (5.9d). We show that $(u, \lambda) \in V \times \Lambda_C$ solves Problem 6.1. Since $u \in V$ and the bulk equation (5.9a) holds with $f \in L^2(\Omega; \mathbb{R}^d)$, we have $\sigma(u) \in H(\mathrm{div}; \Omega)$. Because $|\Gamma_N| = 0$, the load functional reduces to

$$\ell(v) = \int_\Omega f \cdot v \,\mathrm{d}x \qquad \forall\, v \in V.$$

The identity (5.5) for the local normal trace on Γ_C is

$$\langle \sigma_C(u)\mathrm{n},\, \gamma_C v \rangle_{\Gamma_C} = a(u,v) - \ell(v) \qquad \forall\, v \in V. \tag{6.2}$$

Using the decomposition $\sigma_C(u)\mathrm{n} = \sigma_{C,\mathrm{n}}(u)\mathrm{n} + \sigma_{C,t}(u)$, we further get

$$\langle \sigma_C(u)\mathrm{n},\, \gamma_C v \rangle_{\Gamma_C} = \langle \sigma_{C,\mathrm{n}}(u),\, v_\mathrm{n} \rangle_{\Gamma_C} + \langle \sigma_{C,t}(u),\, v_t \rangle_{\Gamma_C}.$$

By (5.9b), the tangential term vanishes, and since $\lambda = \sigma_{C,\mathrm{n}}(u)$ we conclude that

$$a(u,v) - \langle \lambda,\, v_\mathrm{n} \rangle_{\Gamma_C} = \ell(v) \qquad \forall\, v \in V,$$

which is exactly (6.1a). It remains to prove (6.1b). Let $\mu \in \Lambda_C$ be arbitrary. Since (5.9c) implies $u_\mathrm{n} \in W^-_{C,\mathrm{n}}$, and since $\mu \in \Lambda_C$, we have $\langle \mu,\, u_\mathrm{n} \rangle_{\Gamma_C} \geq 0$. Moreover, by (5.9e),

$$\langle \lambda,\, u_\mathrm{n} \rangle_{\Gamma_C} = \langle \sigma_{C,\mathrm{n}}(u),\, u_\mathrm{n} \rangle_{\Gamma_C} = 0.$$

Therefore,

$$\langle \mu - \lambda,\, u_\mathrm{n} \rangle_{\Gamma_C} = \langle \mu,\, u_\mathrm{n} \rangle_{\Gamma_C} - \langle \lambda,\, u_\mathrm{n} \rangle_{\Gamma_C} \geq 0,$$

which proves (6.1b). Hence (u, λ) solves Problem 6.1. Conversely, assume that $(u, \lambda) \in V \times \Lambda_C$ solves Problem 6.1. We first derive the bulk equation. Let $\varphi \in \mathcal{D}(\Omega; \mathbb{R}^d)$. Since $\gamma\varphi = 0$ on Γ, we have $\varphi_{\mathrm{n}} = 0$ on Γ_C. Testing (6.1a) with $v = \varphi$, we obtain

$$a(u, \varphi) = \ell(\varphi) = \int_\Omega f \cdot \varphi \, \mathrm{d}x.$$

Using Proposition 5.2, this becomes

$$\int_\Omega \sigma(u) : \nabla\varphi \, \mathrm{d}x = \int_\Omega f \cdot \varphi \, \mathrm{d}x \qquad \forall\, \varphi \in \mathcal{D}(\Omega; \mathbb{R}^d),$$

which is exactly

$$-\operatorname{div} \sigma(u) = f \qquad \text{in } \mathcal{D}'(\Omega; \mathbb{R}^d).$$

Since $f \in L^2(\Omega; \mathbb{R}^d)$ and $\sigma(u) \in L^2(\Omega; \mathbb{S}^d)$, this implies $\sigma(u) \in H(\operatorname{div}; \Omega)$ and

$$-\operatorname{div} \sigma(u) = f \qquad \text{a.e. in } \Omega.$$

Next, we derive the complementarity relation for λ. Testing (6.1b) with $\mu = 0$ and $\mu = 2\lambda$, we get

$$-\langle \lambda,\, u_{\mathrm{n}} \rangle_{\Gamma_C} \geq 0 \qquad \text{and} \qquad \langle \lambda,\, u_{\mathrm{n}} \rangle_{\Gamma_C} \geq 0,$$

hence

$$\langle \lambda,\, u_{\mathrm{n}} \rangle_{\Gamma_C} = 0. \tag{6.3}$$

We now show that u solves the weak formulation in Problem 4.1. Let $v \in K$ be arbitrary. Then $v_{\mathrm{n}} \in W_{C,\mathrm{n}}^-$. Since $\lambda \in \Lambda_C$, it follows that $\langle \lambda,\, v_{\mathrm{n}} \rangle_{\Gamma_C} \geq 0$. Using (6.1a) and (6.3), we obtain

$$a(u, v-u) - \ell(v-u) = \langle \lambda,\, (v-u)_{\mathrm{n}} \rangle_{\Gamma_C} = \langle \lambda,\, v_{\mathrm{n}} \rangle_{\Gamma_C} - \langle \lambda,\, u_{\mathrm{n}} \rangle_{\Gamma_C} = \langle \lambda,\, v_{\mathrm{n}} \rangle_{\Gamma_C} \geq 0.$$

Thus u solves the weak formulation in Problem 4.1. Since we have already shown that $\sigma(u)$ belongs to $H(\operatorname{div}; \Omega)$, the previously established equivalence of weak and hybrid formulation implies that u solves Problem 5.1. As $|\Gamma_N| = 0$, the previously established equivalence of hybrid and KKT formulation shows that u also solves Problem 5.2. It remains to identify the multiplier λ. Since u solves Problem 5.2, we know in particular that

$$\sigma_{C,t}(u) = 0 \qquad \text{in } W_{C,t}'.$$

On the other hand, because $|\Gamma_N| = 0$ and (5.9a) holds, the local Green formula (6.2) gives

$$a(u, v) - \ell(v) = \langle \sigma_C(u)\mathrm{n},\, \gamma_C v \rangle_{\Gamma_C} \qquad \forall\, v \in V.$$

Comparing (6.2) with (6.1a), we obtain

$$\langle \sigma_C(u)\mathrm{n},\, \gamma_C v \rangle_{\Gamma_C} = \langle \lambda,\, v_{\mathrm{n}} \rangle_{\Gamma_C} \qquad \forall\, v \in V.$$

Using again the decomposition $\sigma_C(u)\mathrm{n} = \sigma_{C,\mathrm{n}}(u)\mathrm{n} + \sigma_{C,t}(u)$ and $\sigma_{C,t}(u) = 0$ on Γ_C, this becomes

$$\langle \sigma_{C,\mathrm{n}}(u),\, v_{\mathrm{n}} \rangle_{\Gamma_C} = \langle \lambda,\, v_{\mathrm{n}} \rangle_{\Gamma_C} \qquad \forall\, v \in V.$$

Now let $\eta \in W_{C,\mathrm{n}}$ be arbitrary. By definition of $W_{C,\mathrm{n}}$, there exists $v \in V$ such that $v_{\mathrm{n}} = \eta$. Hence

$$\langle \sigma_{C,\mathrm{n}}(u),\, \eta \rangle_{\Gamma_C} = \langle \lambda,\, \eta \rangle_{\Gamma_C} \qquad \forall\, \eta \in W_{C,\mathrm{n}},$$

and therefore $\lambda = \sigma_{C,\mathrm{n}}(u)$, which completes the proof. ∎

7 Conclusion and Outlook

In this report, we studied Signorini's problem for a linearly elastic body in unilateral, frictionless contact with a rigid foundation. Starting from the geometrical and physical setup, we introduced the relevant Sobolev and trace theoretic framework and formulated the problem first in terms of admissible displacements and elastic energy. From there, we derived and compared several equivalent formulations, namely the *energy formulation*, the *weak formulation* as a variational inequality, stronger formulations involving normal traces and contact conditions, and finally a *mixed formulation* with a Lagrange multiplier representing the normal contact force. In this way, the report aimed to make precise how the different viewpoints on Signorini's problem are related and how the contact conditions can be recovered in increasingly explicit form.

At the same time, the present work only treats the frictionless case. A natural extension is to include tangential contact effects, i.e. friction, on the contact boundary. In the book [1], this is pursued first through the *Tresca friction model* and afterwards through *Coulomb friction*. From the analytical point of view, this leads to significantly more difficult problems. While the frictionless Signorini problem still fits into the relatively well-structured setting of a variational inequality of the first kind, friction introduces additional nonlinearities and, in the Coulomb case, even a quasi-variational structure. Thus, the frictionless setting considered here may be viewed as the basic model from which more realistic, but also more delicate, contact models can be developed.

Another natural continuation concerns numerical approximation. Since the Signorini problem already admits a weak and a mixed formulation, it fits naturally into the framework of *finite element methods*. In particular, the variational inequality formulation provides the starting point for conforming finite element discretizations, while the mixed formulation is well suited for methods involving *Lagrange multipliers* for the contact force. Beyond this, the book also discusses *Nitsche-type methods* and *augmented Lagrangian techniques* as alternative approaches.

A References

[1] Franz Chouly, Patrick Hild, and Yves Renard. *Finite Element Approximation of Contact and Friction in Elasticity*, volume 48 of *Advances in Continuum Mechanics*. Springer International Publishing, 2023.

B List of Figures